

Quantitative Forest Genetics and Genomics: from linear mixed models to AI-powered prediction

Twenty years of breeding value prediction through increasingly biologically informed models

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Tecnología Agropecuaria



Ministerio
de Economía
República Argentina

Secretaría de Agricultura,
Ganadería y Pesca

General goal

To present the evolution of predictive models in forest tree breeding over the last two decades, highlighting how the progressive integration of biological information -from pedigree and genomics to enviromics, multi-omics and AI- has transformed breeding value prediction.

Presentation Roadmap



PART I

Foundations of Prediction

From classical models to genomics

- 1 The Big Picture
- 2 Spatial & Competition Models
- 3 From Pedigree to Genomics



PART II

Making Genomic Selection Operational

From genomic prediction to real breeding programs

- 4 Operational Genomic Selection
- 5 Training Populations & Prediction Across Generations
- 6 Identifying the Right Trees



PART III

Increasing Biological Complexity

From correlations to biological mechanisms

- 7 Multi-trait Prediction
- 8 From Correlation to Causation
- 9 Machine Learning for Genomic Prediction



PART IV

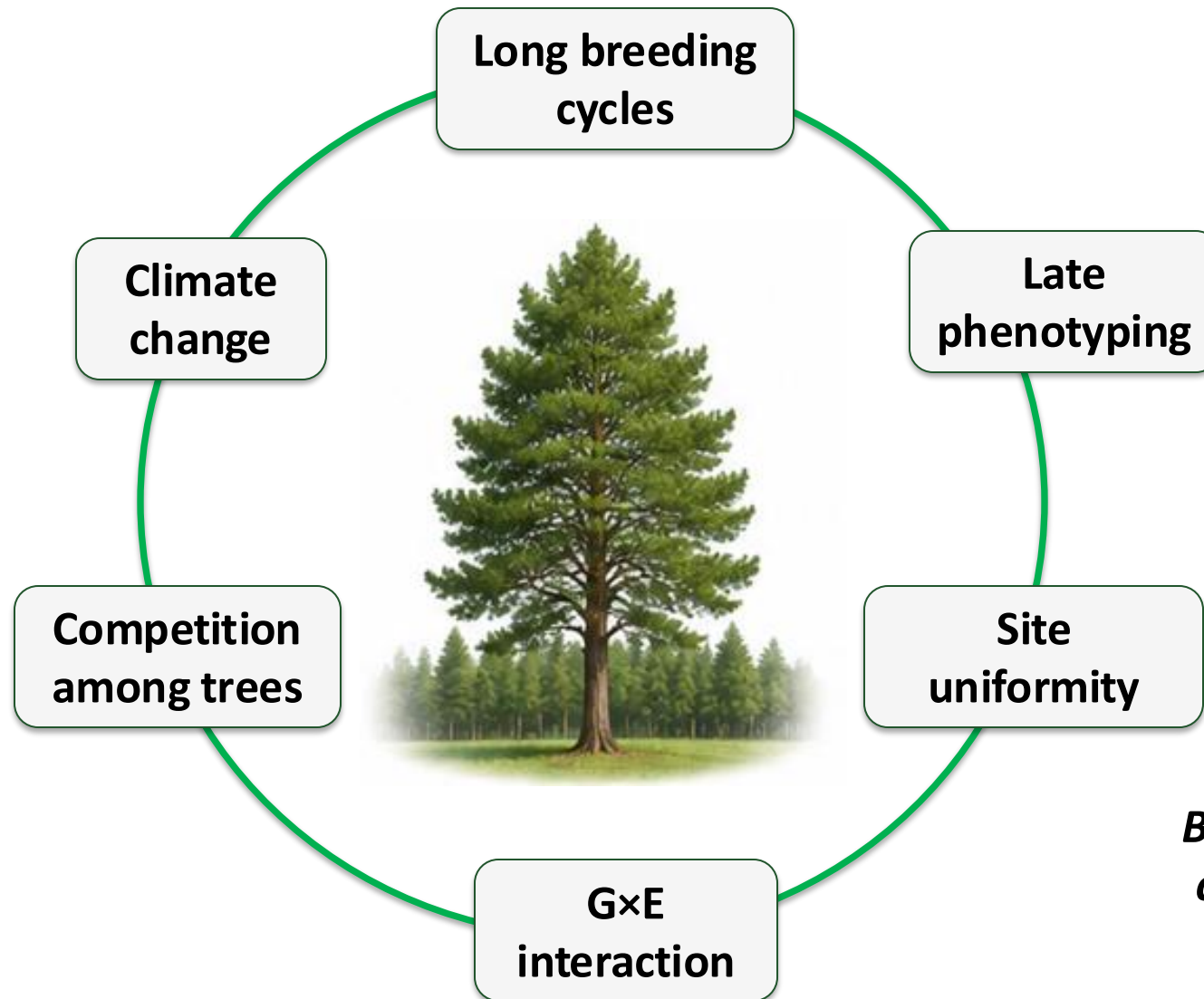
Integrating Environment and Biology

Environment, multi-omics and future systems

- 10 Prediction Across Environments
- 11 Environmental & Biological Covariates
- 12 Looking ahead in forest tree breeding

Why is forest tree breeding so challenging?

Forest tree breeding is one of the most difficult prediction problems in quantitative genetics



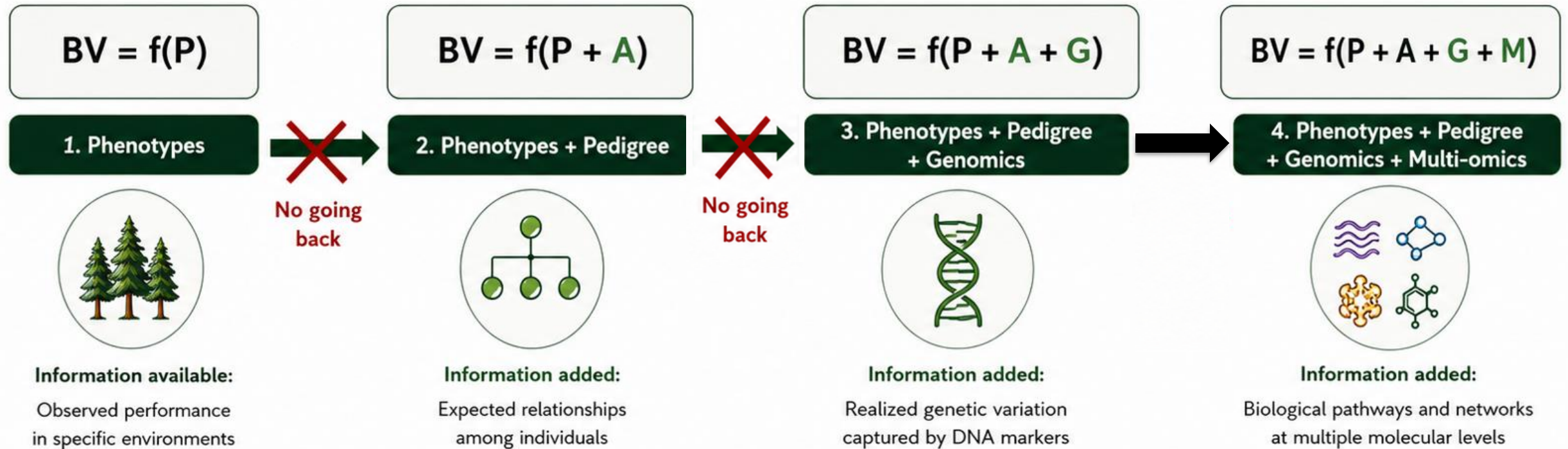
Key challenges

- Trial size: experimental units - area.
- Multi-site field trials
- Multi-trait evaluation
- Repeated measurements over time
- Multi-generations and populations

Breeding value prediction remains a major challenge in forestry genetics, driving the evolution of predictive models.

Progressive integration of biological information

Over the last decades, forest tree breeders have progressively incorporated new sources of biological information into their prediction models.



Progressive integration of biological information

What limitation was each method trying to solve?

Looking back, every methodological advance has emerged as a response to a biological limitation.

Limitation	Methodological advance
Environmental heterogeneity	Spatial models
Tree interactions	Competition models
Pedigree-based expected relationships	Genomics
Independent traits and environments	Multi-trait & multi-environment G×E models
Correlation only	Causal models
Incomplete biological interpretation	Multi-omics
Linear assumptions	Machine Learning

Central hypothesis

**Better predictions emerge as
models progressively capture
biological reality**

A broader trend across breeding and biology

Trends in
Plant Science



Opinion
Better breeding leveraging more biology

Owen M. Powell ^{1,2,3,*}, Lee Hickey ^{1,2,3}, Shunichiro Tomura ^{1,2,3}, and Mark Cooper ^{1,2,3}

Climate-driven variability is reducing our ability to accurately predict crop performance across environments, limiting genetic gain in breeding programs. Sustained progress requires predictive frameworks that capture plant–environment interactions across diverse genetics and management conditions. Integrating mechanistic insights from plant science into predictive models offers a path to improve the accuracy, precision, and interpretability of breeding grounds under uncertainty, precision, and interpretability of emerging hierarchical genome–environment interactions and outline how they can be leveraged within breeding programs to evaluate how biological knowledge informs predictions across target environments and supports long-term genetic gain.

Highlights

Increasing climate-driven variability of cropping systems is testing the limits of current statistical models to predict crop performance across environments.

To improve long-term genetic gain of breeding programs, prediction models must capture the complex interactions across diverse genetic backgrounds, environmental conditions and management practices.



Molecular Plant
Opinion

Biology-driven genebank pangenetics meet
AI-powered predictions to transform crop breeding

The modern era of crop improvement is characterized by two distinct paths: (i) biology-driven molecular genetics focused on dissecting the mechanisms underlying complex traits by a reductionist approach and (ii) data-driven quantitative genetics, which prioritizes accurate prediction of final trait performance, such as yield, without necessarily clarifying the underlying biological mechanisms. These approaches, one statistical association, have each made significant strides, yet with limited overlap. However, their conjunction represents an emerging opportunity: integrating the two ideal crops? Following patterns or understanding the processes that shape them?

We argue that new opportunities for crop improvement lie in striking a balance—combining the statistical power and scalability of quantitative genetics with causal understanding derived from molecular genetics (Figure 1). But how? Here, we discuss progress in molecular and quantitative genetics and envision a future where these can be integrated.



Cell
Leading Edge



Perspective
Why machines don't speak biology:
Toward native biological language models

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<https://doi.org/10.1016/j.cell.2025.07.003>

SUMMARY

Recent advances in AI inspire visions of universal models of biology. Yet living systems are evolved, emergent processes whose behaviors cannot be inferred from their parts alone. We propose grounding AI in canonical biological processes, constructing data-driven world models with explicit mechanistic links across molecules, cells, and their dynamics in space and time.

Common message across disciplines



Better
biological
representation



Better
prediction



Better breeding
decisions

PART I: Spatial & Competition Models

The first limitation: heterogeneous field conditions

Cappa et al. (Sil Gen, 2015) – Tsuga heterophylla

Forest trials are **rarely** environmentally homogeneous.



Soil fertility



Moisture



Topography



Previous land use
or management



These factors create spatial
variation in tree growth
across the trial.

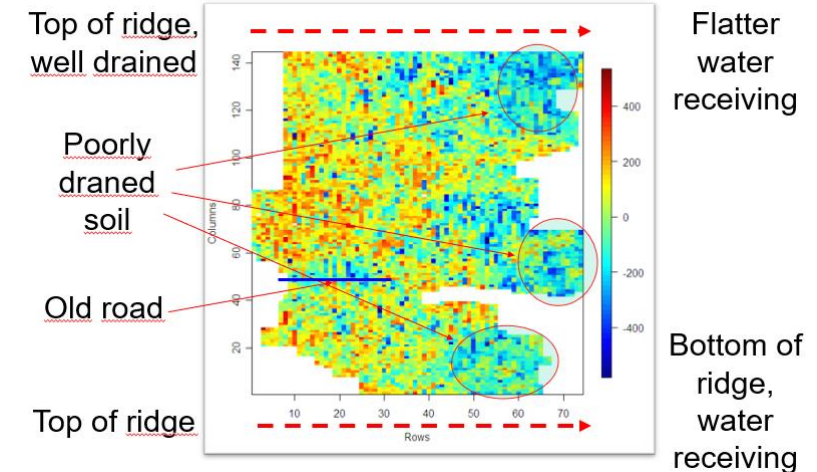
Example of a forest trial



Trial showing environmental gradients



Spatial variation (example)

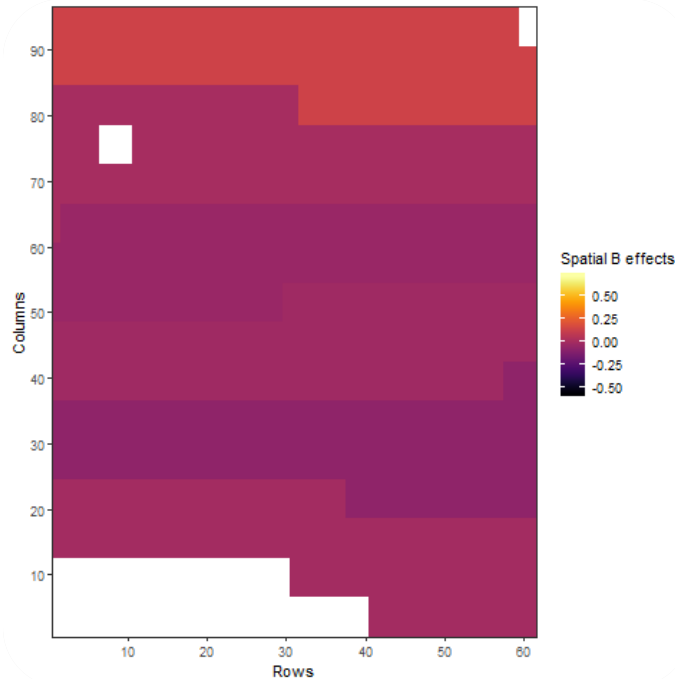


Ignoring environmental heterogeneity can **bias** the estimation of genetic parameters and the prediction of **breeding values** (i.e., BLUP of BV).

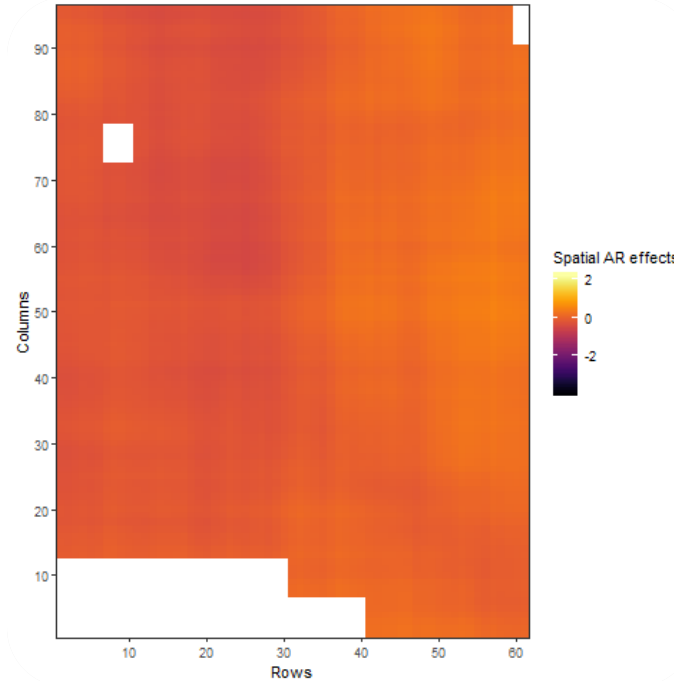
Traditional solution: experimental design

“A Priori” designs to control environmental heterogeneity: RCB or IBD

Randomized complete blocks



Spatial effects



Limitations



Fixed block boundaries



Cannot capture continuous spatial trends



Environmental pattern unknown before establishment

Choosing an optimal design is often difficult or impossible.



Models must learn spatial patterns from the data, “a posteriori”.

Spatial models

Spatial models

Cappa & Cantet (CJFR, 2007) – E. globulus

Several spatial approaches have been proposed to model environmental variation in field trials.



Post-blocking

Optimized incomplete blocks fitted after trial establishment.



AR(1)

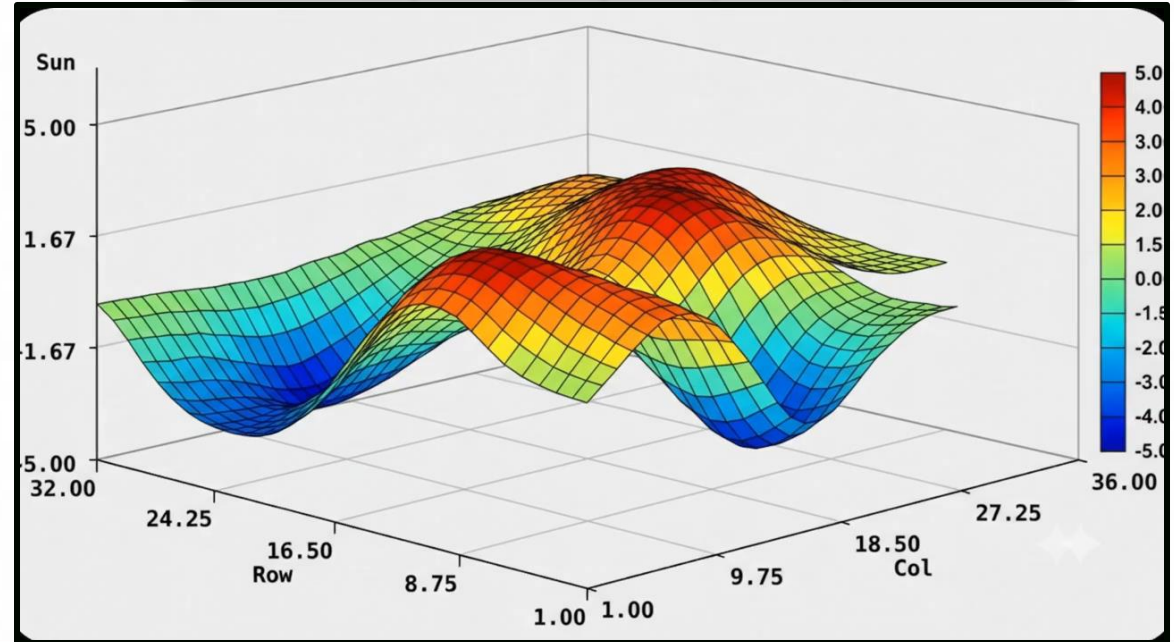
Autoregressive structures for rows and/or columns.



Tensor product B-splines

Two-dimensional smoothing approach based on tensor products of B-spline bases.

Example: spatial surface estimated with tensor-product B-splines (8 knots)



1

Flexible modeling of large- and small-scale variation

Tensor-product B-splines within mixed models capture complex environmental patterns in two dimensions.

2

Improved breeding value prediction

Spatial models reduce environmental noise, increasing heritability, accuracy of breeding values and expected genetic gain.

3

Easily implemented in operational tree breeding

The methodology is computationally efficient and can be incorporated into routine genetic evaluation programs.

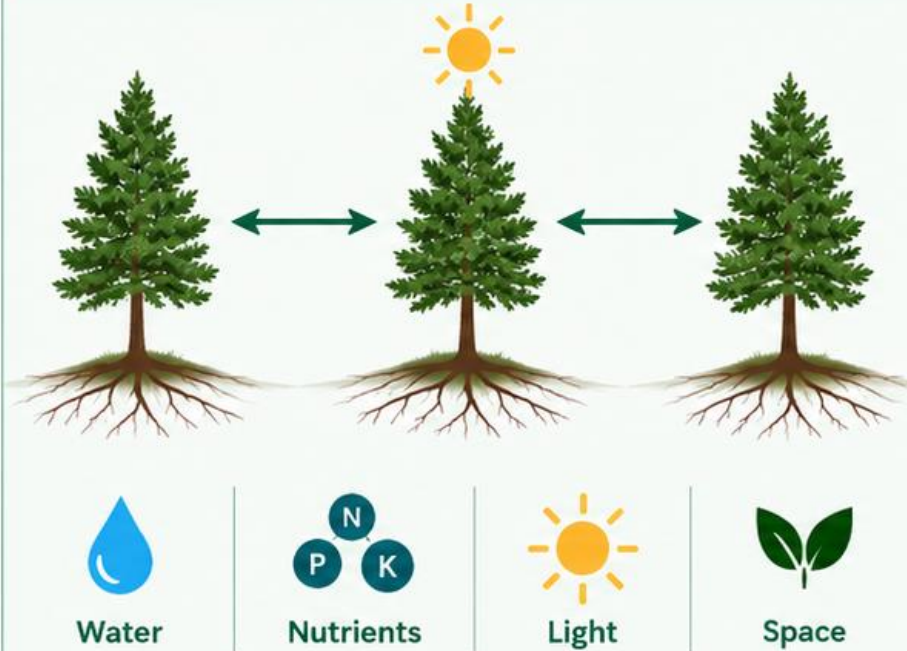


Key message

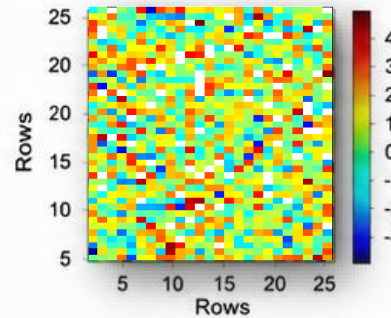
Spatial models were the first major step toward incorporating biological realism into breeding value prediction.

The second limitation: trees do not grow in isolation

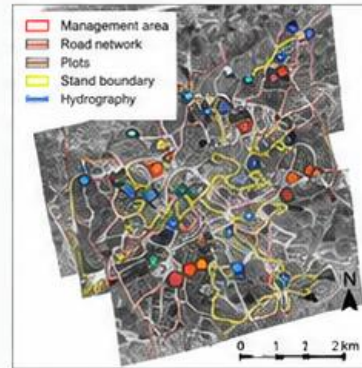
What drives competition?



Evidence from the field

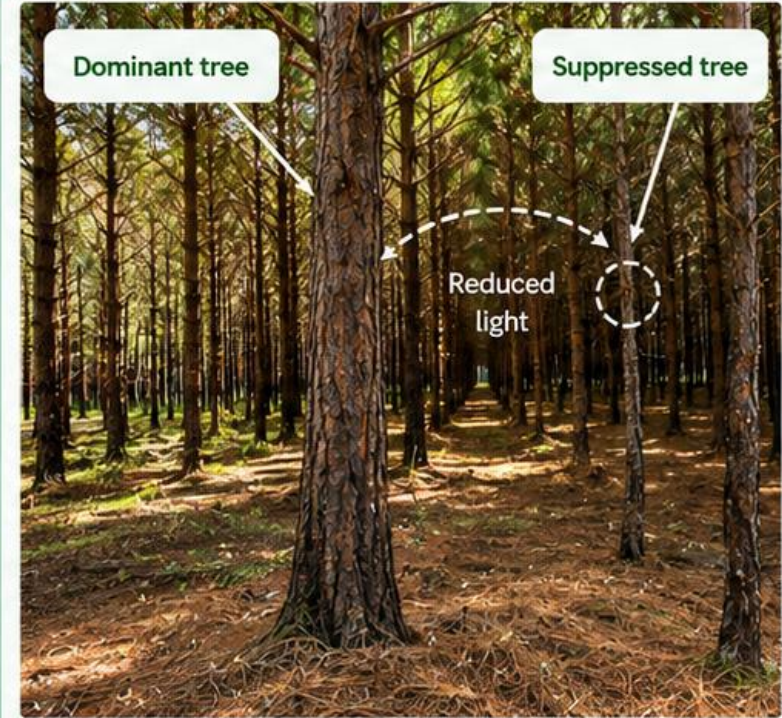


Spatial pattern of residuals DBH in the *Pinus taeda* L.



Aerial view of the root system of a hydraulically excavated clone of *Quercus pyrenaica*.

Competition in the field



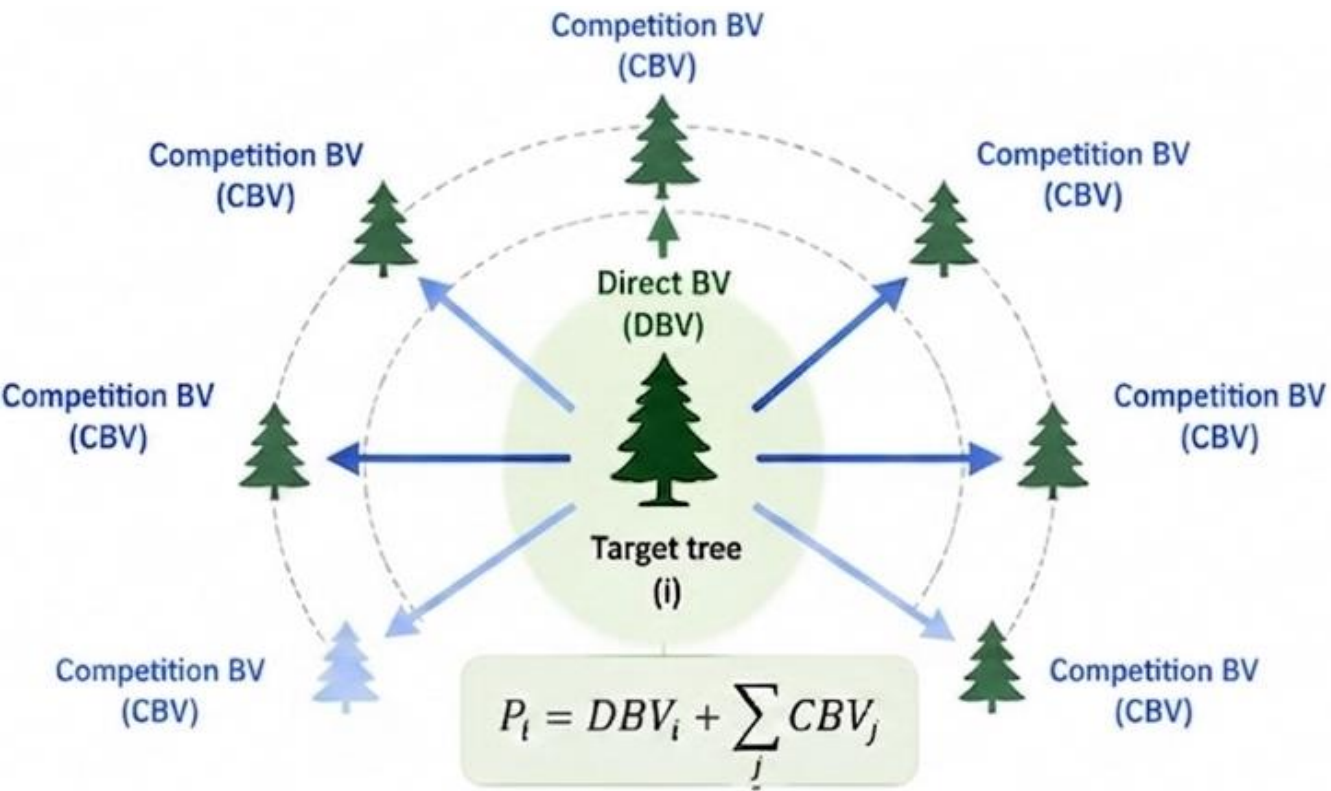
1 Competition is caused by genetic and environmental sources

2 Ignoring competition effects can *bias* the estimation of genetic parameters and the prediction of breeding values (i.e., BLUP of BV).

Competition model

Cappa & Cantet (Sil Gen, 2008) – *P. taeda*

To account for competition, the phenotype of an individual can be modeled as a combination of two breeding values



Direct breeding value (DBV)

- affects its own phenotype.



Competition breeding value (CBV)

- affects neighbouring trees.



Intensity of competition (IC): modelled as an inverse function of neighbor distance and number of competing trees.

Nearest tree



Farther tree



Distant tree

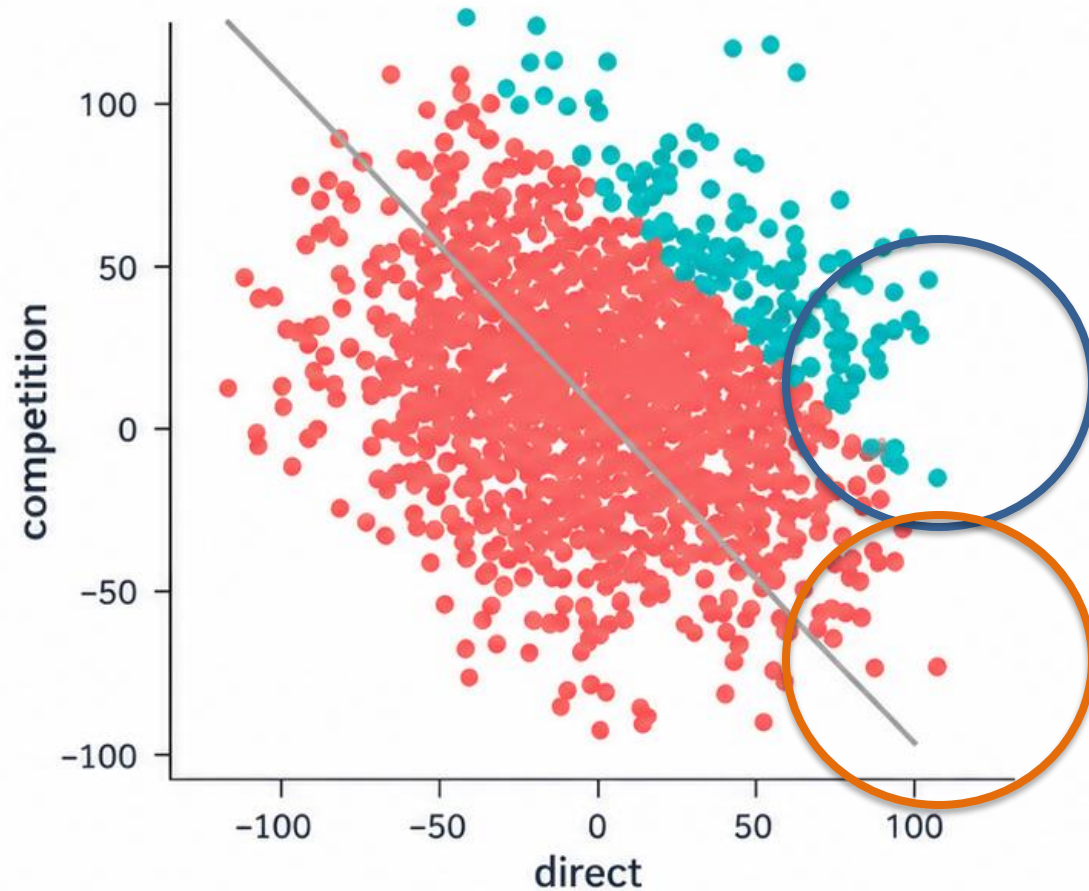


Influence decreases with distance

Competition model: beyond individual performance

Cappa & Cantet (Sil Gen, 2008) – P. taeda

Competition models distinguish direct and competition breeding values for selection decisions.



Traditional selection

- Selects trees with high Direct BV only.
- Tends to favour highly competitive (genotypes that suppress neighbours).



Competition-aware selection

- Selects trees using Direct BV + Competition BV.
- Identifies **productive** and **less competitive** genotypes.

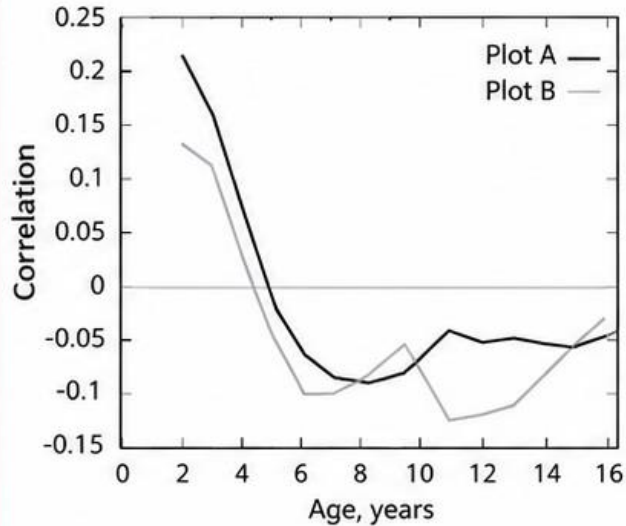


Selection shifts from maximizing individual growth to maximizing **stand productivity**.



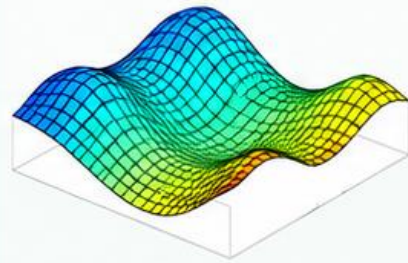
Joint modeling of spatial and competition effects

Cappa et al. (TGG, 2015) – *P. taeda*
Cappa et al. (TGG, 2016) – *Pseudotsuga menziesii*

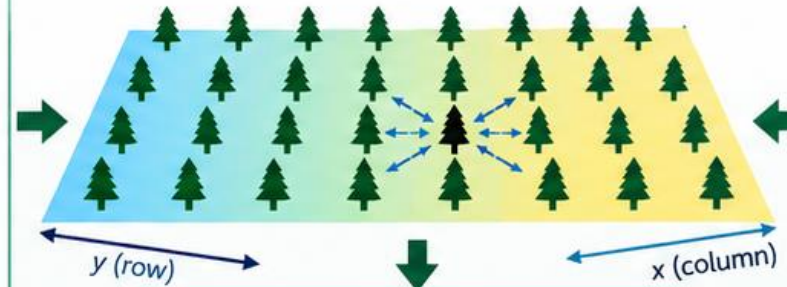


Spatial variation

Environmental heterogeneity modelled with a two-dimensional smoothing surface.



Forest genetic trial

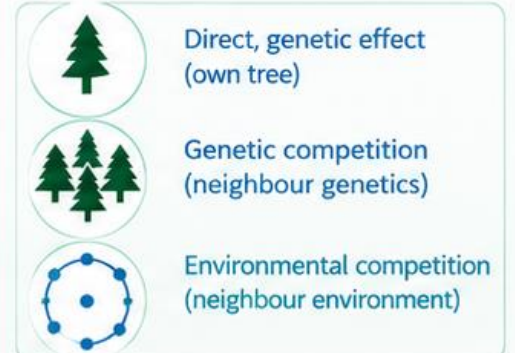


CSM

Competition + Spatial Model
Accounts for both spatial variation and competition (genetic and environmental).

Competition

Direct, genetic and environmental competition effects from neighbouring trees.



WHAT DID THE JOINT MODEL REVEAL?

1

Ignoring competition → overestimates environmental heterogeneity



Higher spatial variance when competition is ignored.

2

Ignoring environmental heterogeneity → underestimates competition effects



Underestimated direct-competition genetic correlation and environmental competition variance.

3

High BV correlations ≠ identical selection



The top 5% ranked trees changed across models (TM-CSM).



TAKE-HOME MESSAGE:

Spatial heterogeneity and competition coexist: joint modeling leads to realistic genetic evaluations and better selections.

PART I: Pedigree to Genomics

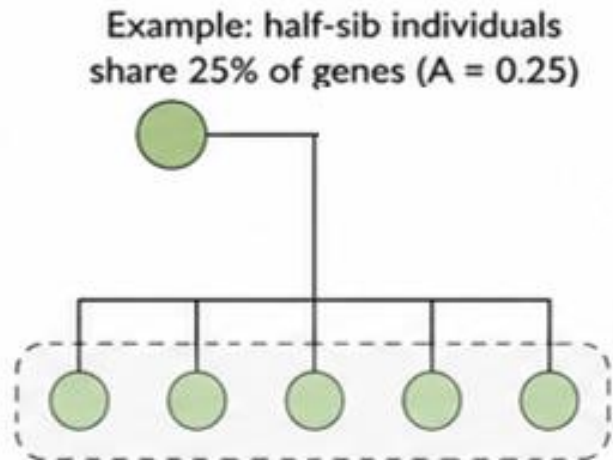
... And the genome became part of the mixed models.

The third limitation: pedigree-based expected relationships

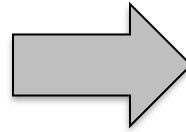
Cappa et al. (Mol Breed, 2016) – *E. grandis*

EXPECTED RELATIONSHIPS from pedigree

- Contains only expected relationships.
- Ignore historical relatedness.

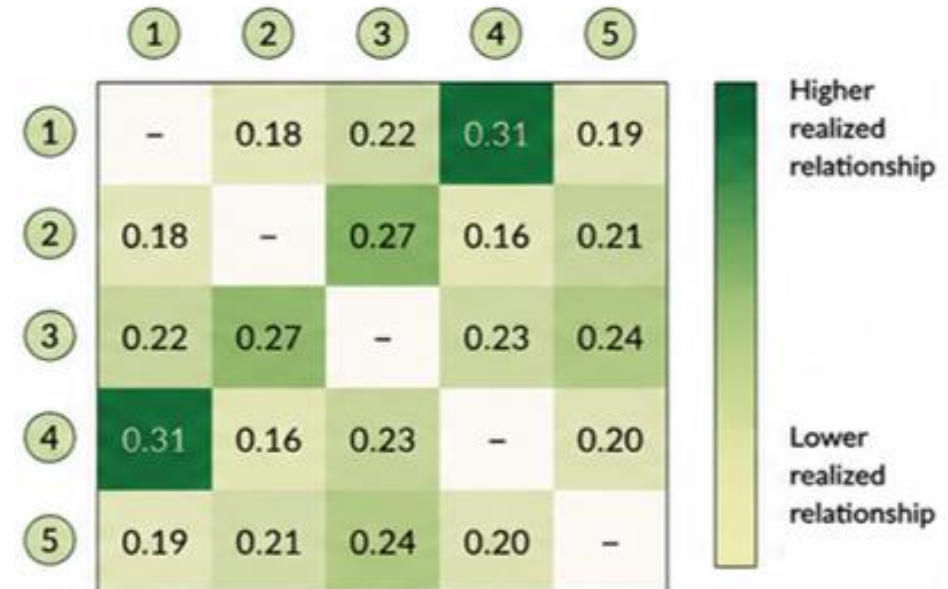


All half-sib pairs have
the same expected relationship:
 $A = 0.25$



REALIZED RELATIONSHIPS estimated with molecular markers

Marker data capture the actual proportion
of genome



Mendelian sampling makes every pair of
relatives genetically unique.



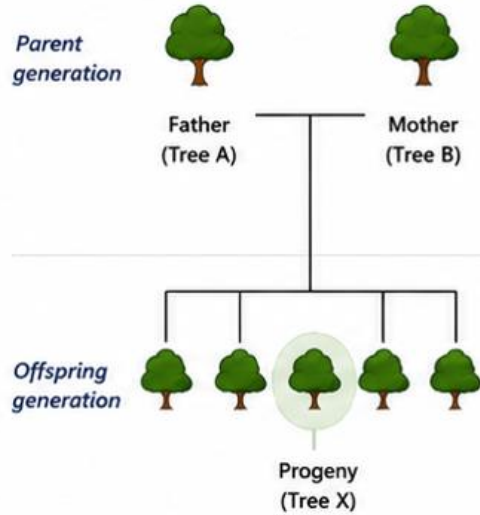
Key message

Improve the accuracy of genetic parameter estimates and
breeding value prediction.

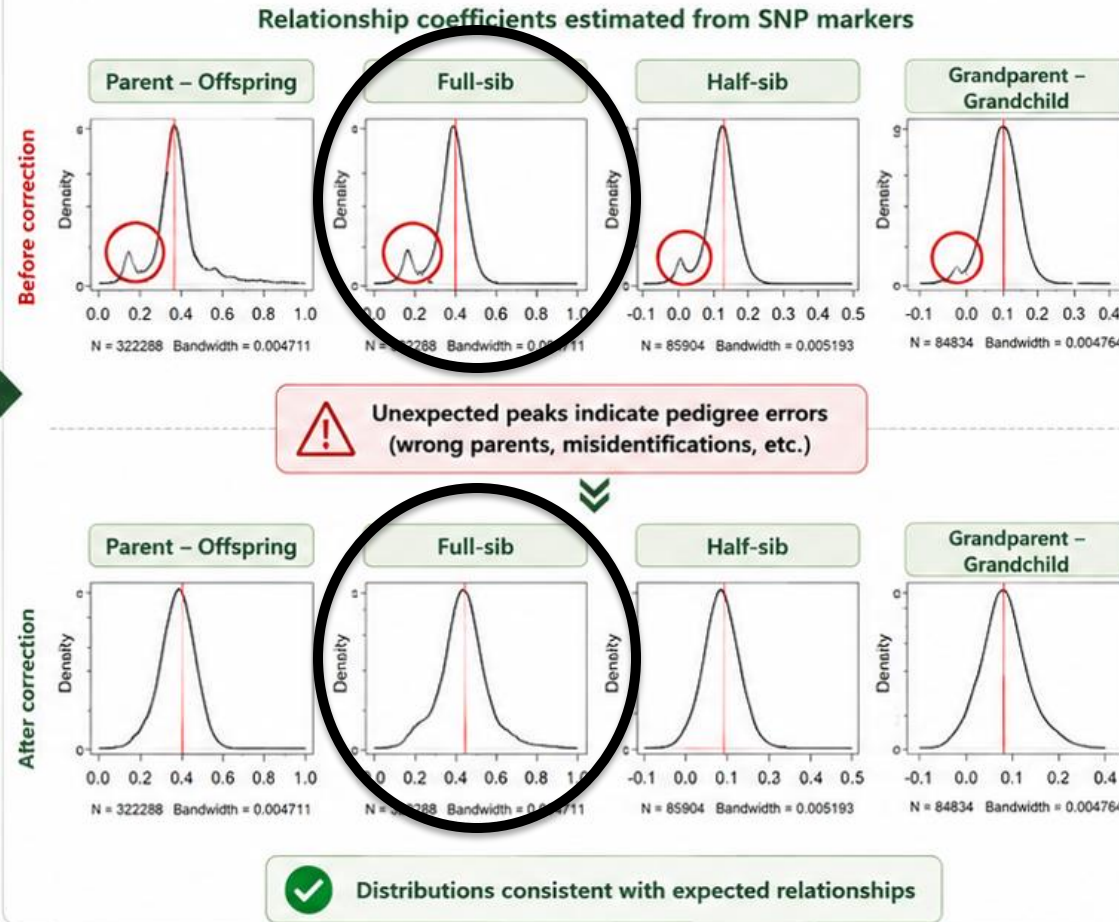


Using genomic data to validate and correct pedigrees

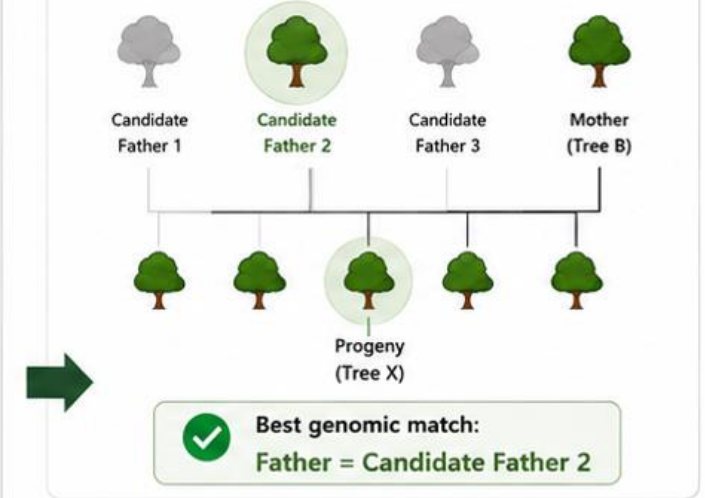
1. Pedigree says...



2. Genomic relationships reveal inconsistencies



3. Pedigree correction



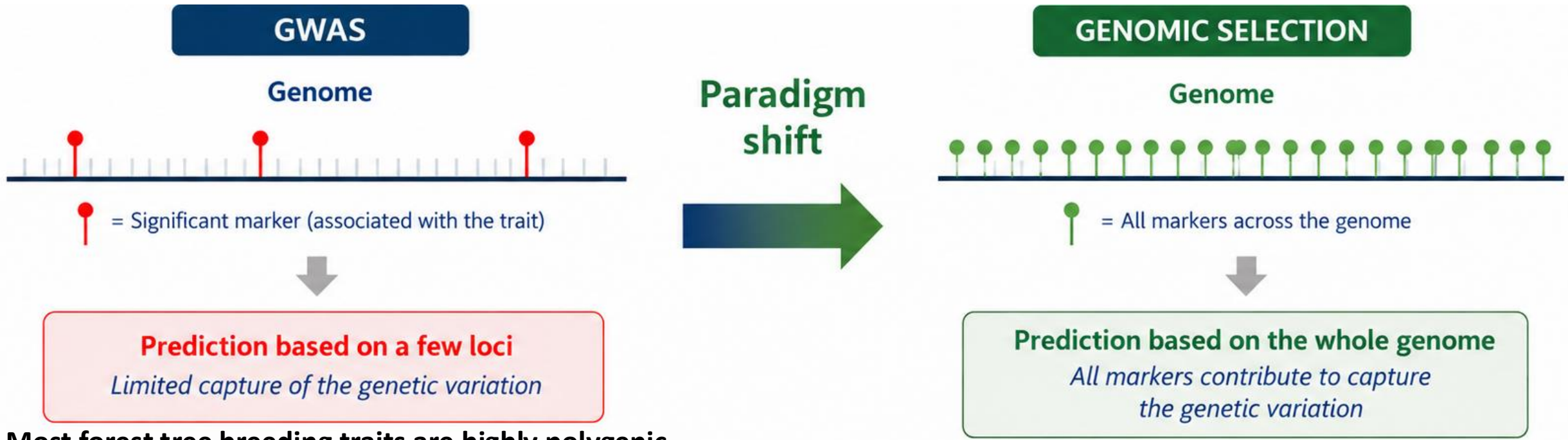
Muñoz et al. (Crop Science, 2014)



Genomic information improves pedigree accuracy by identifying and correcting relationship errors.

From Gene Discovery to Genome-wide Prediction

Cappa et al. (Plos One, 2013) – *E. globulus*



Most forest tree breeding traits are highly polygenic.

GWAS performs best for oligogenic traits,
Shalizi et al. (G3, 2021) – P. taeda – fusiform rust



The question changed from “Which genes matter?” to “How well can we predict?”



GWAS
Gene discovery



Genomic Selection
Genome-wide prediction

How genomic selection works

Grattapaglia et al. (Frontiers in Plant Sci, 2018)

A. GENOMIC MODEL DEVELOPMENT AND UPDATING

GENOTYPES

SNP1 A/G
SNP2 A/A
SNP3 G/G
SNP4 A/C
.....
SNP50k C/C



PHENOTYPES or BREEDING VALUES

Volume
Stem taper
Wood density
Lignin, Cellulose
Disease score
....

Training Population

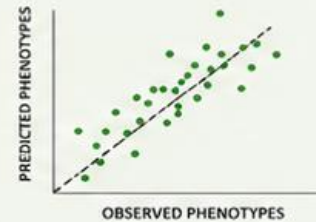
PREDICTIVE MODEL

$$y = Xb + Zh + e$$

MODEL CROSS-VALIDATION



Validation Population



VALIDATED PREDICTIVE MODEL

$$y = Xb + Za + e$$

MODEL UPDATING



Subset of already genotyped seedlings from the previous Genomic Selection generation are planted in field trials. Their phenotypes are measured and used to update the predictive model.

B. GENOMIC SELECTION CYCLE

Genotypes (SNPs)

	1	2	3	4	5	6	7	...	n
1	x	x	x	x	x	x	x	...	x
2	x	x	x	x	x	x		...	x
3			x	x	x	x		...	x
4				x	x	x		...	x
5					x	x	x	...	x
6						x	x	...	x
7							x	...	x
...									

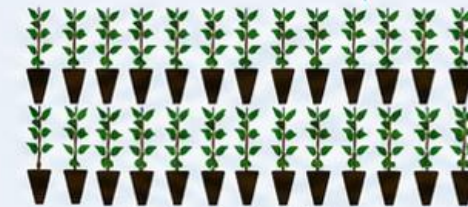
Selected parents



Breeding: mating design or open pollination of elite parents

GENOTYPING OF SELECTION CANDIDATES

SNP1 A/G
SNP2 A/A
SNP3 G/G
SNP4 A/C
.....
SNP50k C/C



Seedling candidates

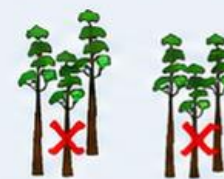
PREDICTIVE MODEL

$$y = Xb + Za + e$$

GENOMIC SELECTION



Top genomically ranked individuals



Clonal trial for field validation

GEGV

GEBV



Flower induction or top grafting

RECURRENT BREEDING CYCLE

The impact of genomic selection in forest tree breeding

Google Scholar

"genomic selection" and "forest" and "tree"

Articles

About 1,080 results (0.13 sec)

Any time

Since 2025

Since 2024

Since 2022

Since 2020

Since 2018

Custom range...

1900 - 2025

Search

Sort by relevance

Sort by date

Any type

Review articles

☐ include patents

☒ include citations

☒ Create alert

"Genomic selection" and "forest" and "tree" – Ask SciSpace AI

SciSpace Literature Review, 2024 - scispace.com

Do a comprehensive literature review for "genomic selection" and "forest" and "tree" in SciSpace and get citation-backed insights from 250M+ scientific research papers

Genomic selection for growth and wood properties in multi-generation hybrid populations of *Populus deltoides*

X Zhou, L Zhang, M Zhang, H Wei, Y Lu - ..., 2023 - academic.oup.com

... a genomic selection reference and breeding population for *P. deltoides* using multi-generational full-sibling progenies, and systematically conducted genomic selection breeding ...

☆ Cited by 28

Related articles

All 7 versions

[PDF] oup.com

Full View

Trends in tree improvement methods: from classical breeding to genomic technologies

S Isik - Annals of... - 2025 - Springer

... Genomic selection and genetic modification of non-model ... The study of annual nuclear tree mortality indicated that forest tree di... of the tree population consistently affect forest health, ...

☆ Cited by 17

Related articles

All 2 versions

The first application of genomic selection for tree morphological traits in *Pinus koraiensis* using markers identified through a genome-wide association study

D Jeong, Y Kang, Y Lim, Y Lee, KS Cheon - ..., 2025 - academic.oup.com

... We present a simulated combined selection scheme utilized for *P. koraiensis*, a three species of important economic and ecological value in Korea and China, utilizing ...

☆ Cited by 1

Related articles

All 2 versions

Search performed on May 20, 2025

Number of publications per year

(Search in Google Scholar: "genomic selection" AND "forest" AND "tree")

Year	Number of publications
antes 2010	160
2010	40
2011	90
2012	120
2013	130
2014	200
2015	200
2016	250
2017	270
2018	320
2019	420
2020	520
2021	620
2022	716
2023	660
2024	776
2025	1,080

More than doubled in just six years

Emerging interest (before 2018)

Rapid adoption and exponential growth

Key message

The science behind genomic selection has advanced rapidly; however, translating that knowledge into operational impact remains challenging.

PART II: Operational Pedigree to Genomics

Genomic prediction became operationally feasible.

Integrating pedigree and genomics

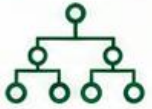
Cappa et al. (IUFRO Forest Tree Breeding Conference, 2014) – *E. grandis*

One of a major practical challenge: thousands of valuable individuals in the pedigree remained non-genotyped.

The process

1

A: expected pedigree relationships



- Based on recorded ancestry
- Captures between-family (parental average) variation

2

G: realized genomic relationships



- Based on DNA markers
- Captures within-family (Mendelian sampling) variation

3

H: integrated relationship structure



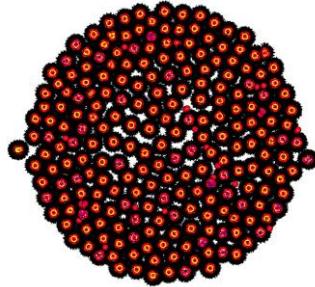
- Combines A and G
- Enables joint evaluation of genotyped and non-genotyped individuals



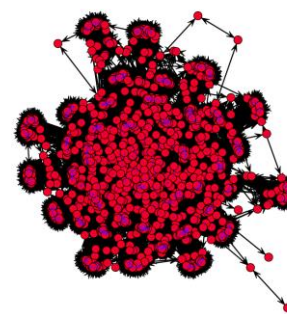
Genotyped and non-genotyped individuals are evaluated jointly.

From A and G to H

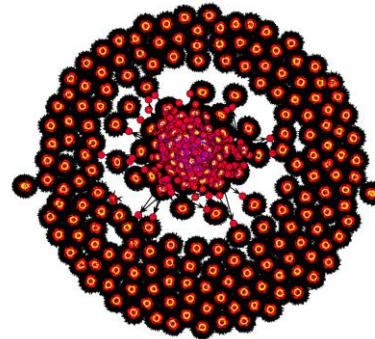
A-matrix
(pedigree)



G-matrix
(genomic)



H-matrix
(pedigree + genomic)



- Non-genotyped
- Genotyped

Mistal et al. 2009; Legarra et al. 2009; Christensen and Lund 2010

MODEL: HBLUP (single-step GBLUP)



Joint use of phenotypic, pedigree and genomic information

$$\underbrace{\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + H^{-1}\alpha \end{bmatrix}}_{\text{Matrix of coefficients}} \underbrace{\begin{bmatrix} \hat{\beta} \\ \hat{g} \end{bmatrix}}_{\text{Solutions}} = \underbrace{\begin{bmatrix} X'y \\ Z'y \end{bmatrix}}_{\text{RHS}}$$

H matrix

$$H = \begin{bmatrix} \text{Var (Non-genotyped parents and offspring)} & \text{Cov (Non-genotyped parenty and offspring, Genotyped offspring)} \\ \text{Cov (Non-genotyped parents and offspring, Genotyped offspring)} & \text{Var (Genotyped offspring)} \end{bmatrix}$$

Inverse of H

$$H^{-1} = A^{-1} + \begin{bmatrix} 0 & 0 \\ 0 & G^{-1} - A_{22}^{-1} \end{bmatrix} \quad \text{Aguilar et al. 2010}$$



This formulation of H^{-1} is computationally efficient.



The H -matrix combines both pedigree-based parental averages (between-family variation) and marker-based Mendelian sampling effects (within-family variation).

Genomic information enhances theoretical accuracy



Key message

Genomic relationships, corrected pedigrees, and ssGBLUP **improve breeding value accuracy.**



Does genomic prediction really work across generations?

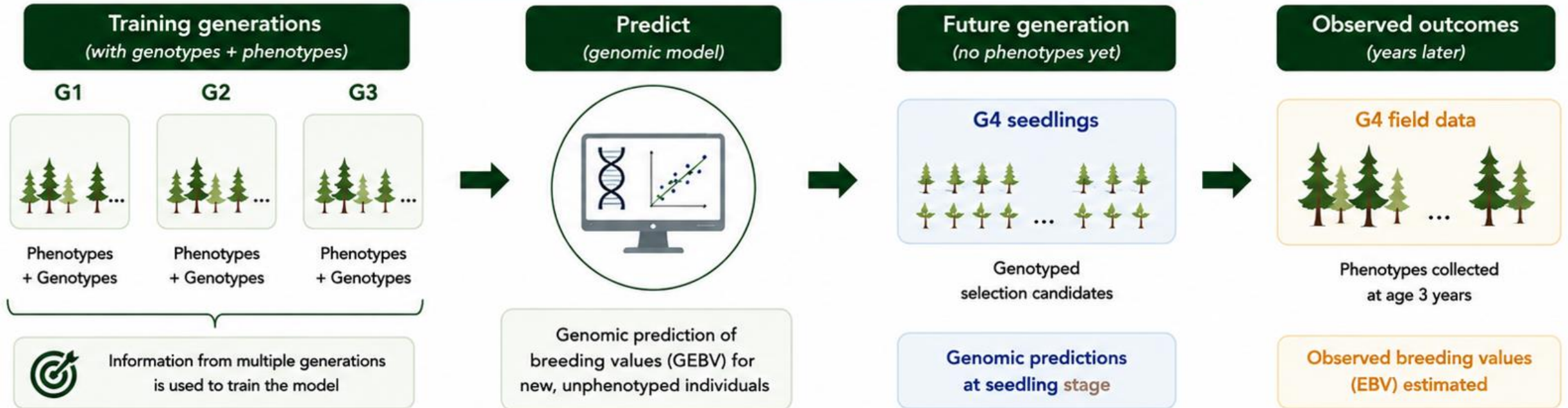
Duarte et al. (Frontiers in Plant Sci, 2024) – *E. grandis*

Cross-validation is useful...

...but does genomic prediction work in future generations?



Accurate assessments of prediction accuracy, can only be made after the completion of the GS cycle



Realized Predictive Ability (RPA)

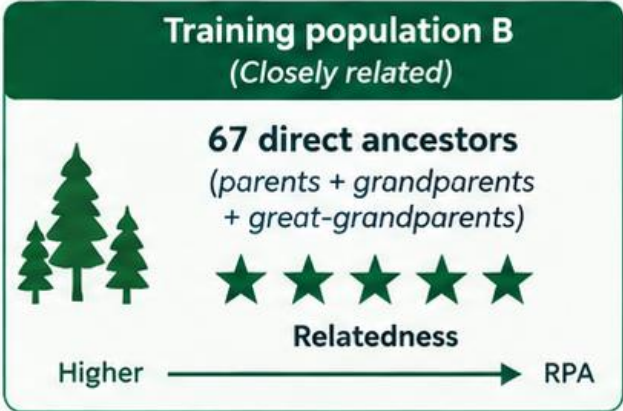
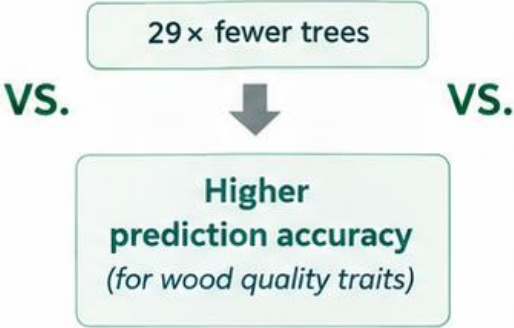
Correlation between genomic predictions (GEBV) in G4 and later-observed breeding values (EBV).

What makes a good training population?

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

Does a larger training population always produce better predictions?

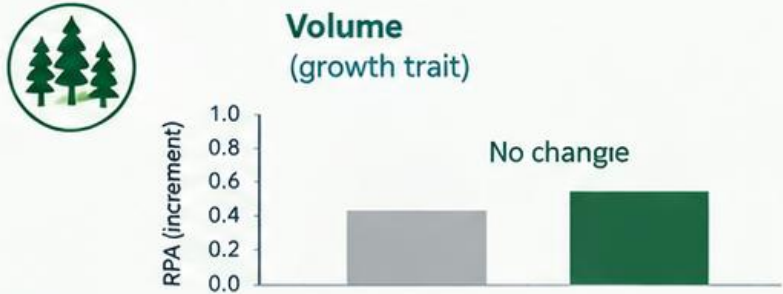
Not necessarily.



Why?

Close relatives share more informative genetic relationships with selection candidates.

Realized Predictive Ability (RPA) across traits and training populations



Key message

High relatedness outperforms population size in training sets.



Realized predictive ability (RPA): different traits, different challenges

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

RPA across traits

	 Volume (VOL) Growth trait	 Wood density (WD) Wood quality trait	 Pulp yield (PY) Wood quality trait
 RPA (individual-tree)	0.11	0.43	0.72
 RPA (between-family)	0.16	0.42	0.75



GS is much more effective for wood quality traits.

RPA (individual-tree): PY (0.72) > WD (0.43) >> VOL (0.11)
RPA (between-family): PY (0.75) > WD (0.42) >> VOL (0.16)

Why is volume (VOL) so difficult to predict?



Lower heritability
Less genetic variation is captured, reducing predictive ability.



Highly polygenic
Many genes with small effects make prediction more challenging.



Stronger dependence on genome-wide LD
Prediction relies more on LD patterns, which may not transfer well.



Family selection is particularly advantageous for difficult-to-predict traits.



Two-stage genomic selection
Select families first → then individuals within top families.
Higher selection intensity and lower genotyping costs.



Key message

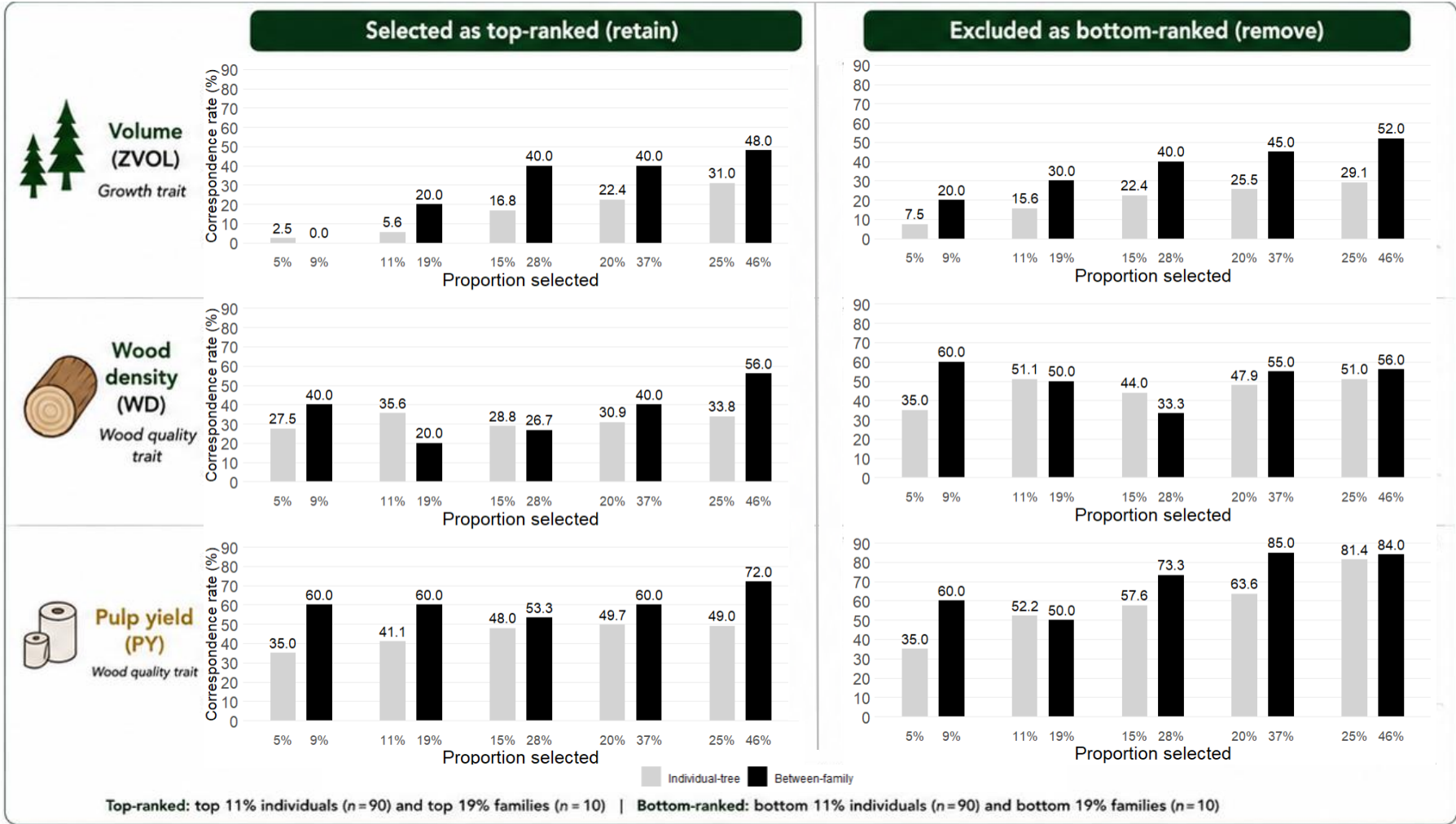
GS works, but performance depends on the trait.



Can genomic selection identify the right trees?

Duarte et al. (Frontiers in Plant Sci, 2024) – *E. grandis*

Correspondence between genomic (GEBV) selection and phenotypic (EBV) selection across selection intensities



Wood quality traits
GS accurately identifies superior trees and families

- ✓ Wood density (WD)
- ✓ Pulp yield (PY)

Correspondence up to **60–70%** even at high selection intensity (5–11%).

Family selection is more reliable than individual-tree selection

- ✓ Consistently higher correspondence at all selection intensities.
- ✓ At 5% selected, ~60% of top-ranked families for WD and PY are captured.

Supports a two-stage genomic selection strategy:
families first, then individuals within top families.

Early culling advantage
GS is even better at removing the worst performers.

- ✓ Higher correspondence for bottom-ranked trees and families.
- ✓ Critical for optimizing resources early in the breeding cycle.



GS is not only useful for choosing best trees.

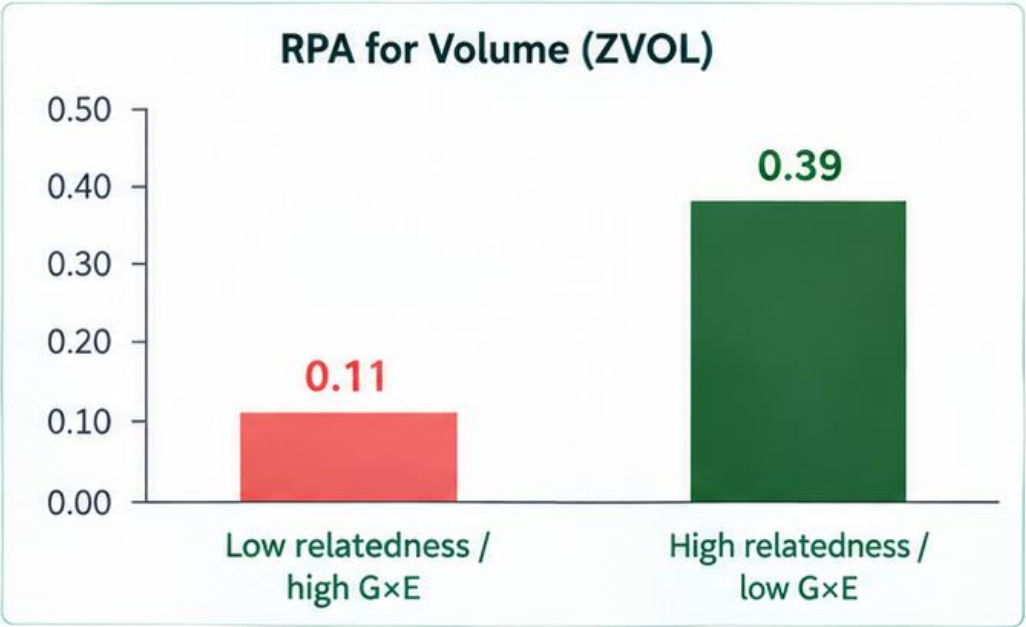


Why is prediction for volume growth so challenging?

Duarte et al. (Frontiers in Plant Sci, 2024) – *E. grandis*

Key training-to-testing scenarios for volume growth (Trial 8 = testing)			
Scenario	Relatedness (traing → testing)	G×E	RPA
 Low relatedness / high G×E	Low	High	0.11
 High relatedness / low G×E	High	Low	0.39

~3x
higher RPA



- 

1 Relatedness matters
Closer training–testing relationships improve RPA.
- 

2 Environmental similarity matters
Low G×E improves prediction transfer across sites.
- 

3 Model and training set matter
ssGBLUP + direct ancestors + diverse training sets improve RPA.



Genomic selection is mature as a **statistical methodology**.

It's operational success depends on understanding the breeding population, the trait, the data and the biological context.



PART III: Multi-trait Biological Integration

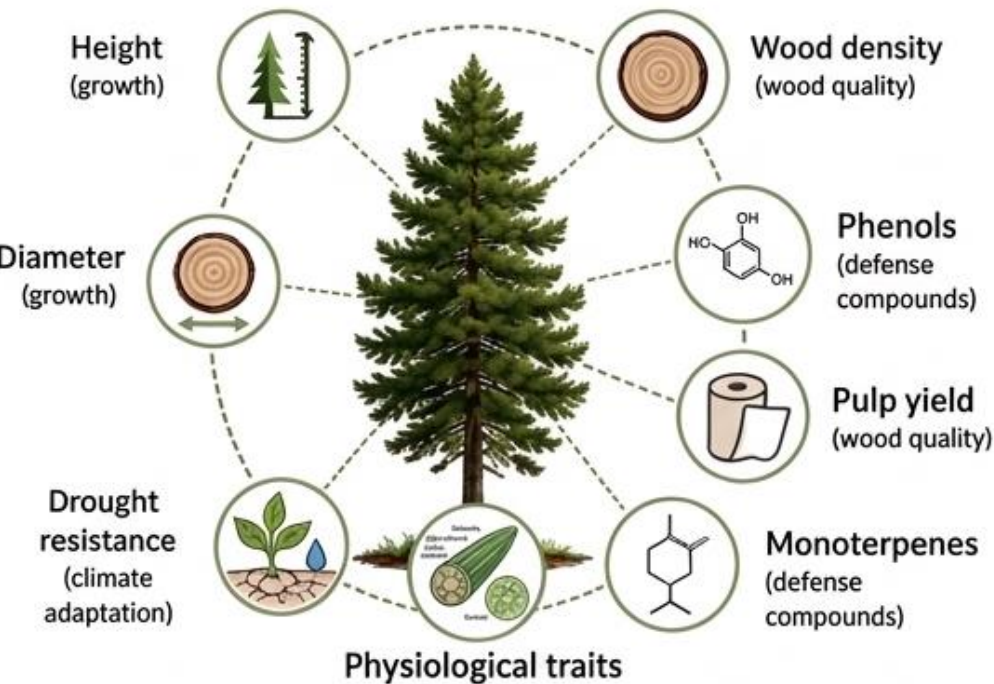
The fourth limitation: traits do not exist independently

Cappa et al. (BMC Gen., 2022) – Lodgepole pine

Traits are not independent: they share biological functions and genetic information.

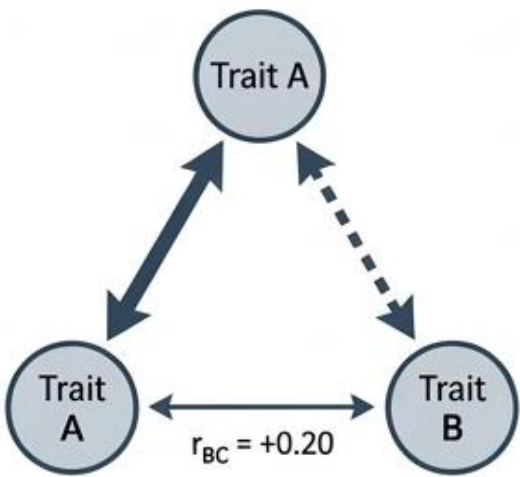
1. BIOLOGICAL SYSTEM

Traits are connected through biology and physiology.



2. GENETIC RELATIONSHIPS

Traits are genetically correlated and share information.



Genetic correlations

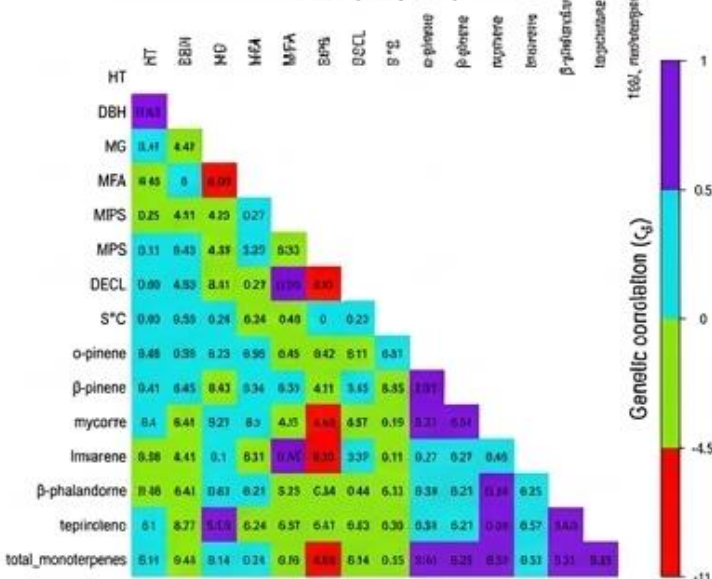
3. INFORMATION SHARING AND PREDICTION



Multi-trait model
Borrow information across genetically correlated traits.

Traits are genetically correlated

JUDY (example trial)



Genetic correlations create opportunities to borrow information across traits.

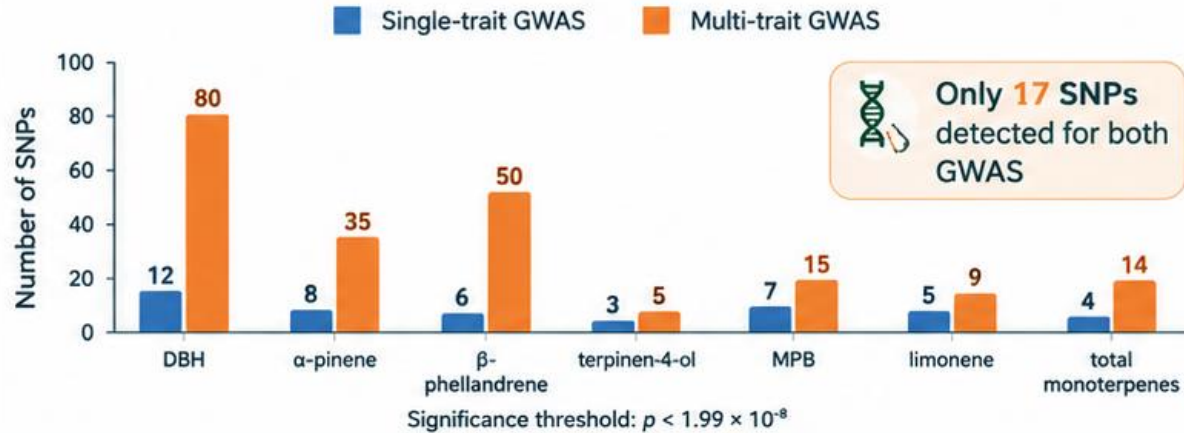
Multi-trait models: impact on GWAS power and accuracy

Cappa et al. (BMC Gen., 2022) – Lodgepole pine

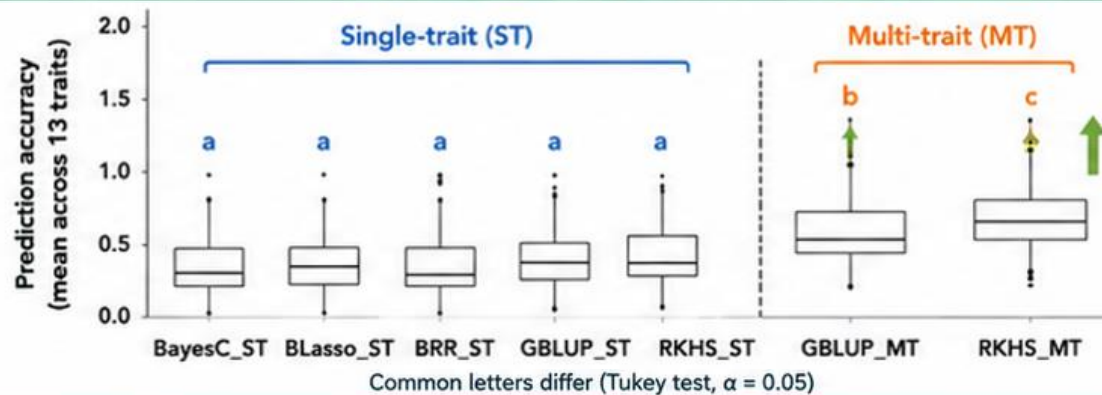
Jurcic et al. (TGG, 2023) – E. grandis



More QTL detected: increased GWAS power



Higher prediction accuracy



Predicting traits that are difficult or expensive to measure



Challenge

Wood-quality traits are expensive and often sparsely recorded.



Multi-trait strategy

Borrow information from easily measured growth traits.



Result

Higher predictive accuracy for difficult-to-measure traits.



Up to +17.7% in predictive accuracy

Multi-trait models can *extend genomic prediction* to traits that are costly or *difficult* to measure.



Key message

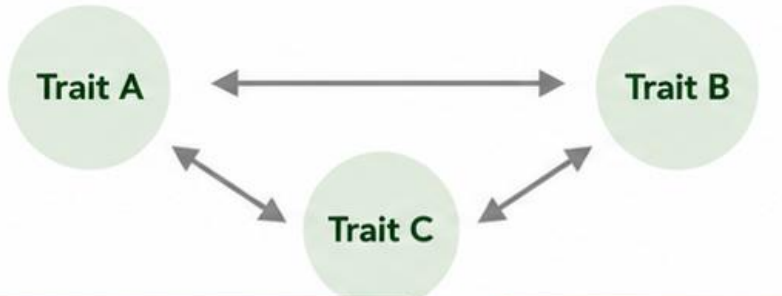
Borrowing information across traits enhances biological discovery, improves prediction, and enables prediction of difficult or expensive recorded traits.



PART III: From Correlation to Causation

The fifth limitation: correlation is not causation

Traits can be **correlated**... but correlation does not tell us **why**.



? What generates these **correlations**?

CORRELATION (what we usually see)

A diagram showing three light green circles labeled 'Trait A', 'Trait B', and 'Trait C'. Dashed lines connect Trait A to Trait B, Trait A to Trait C, and Trait B to Trait C, representing correlation.

We only observe that traits **move together**.

VS.

CAUSATION (what we want to uncover)

A diagram showing three light green circles labeled 'Trait A', 'Trait B', and 'Trait C'. Solid arrows point from Trait A to Trait B, Trait A to Trait C, and Trait B to Trait C, representing causal relationships.

One trait influences another through **direct or indirect effects**.

Why the same correlation can arise

Shared genetics
(pleiotropy / linkage)
Traits are correlated because of their shared genetic basis.

Biological regulation
Traits are correlated because one affects another through biological pathways.

Correlation ≠ Causation
They do not reveal the underlying mechanisms that generate those associations.

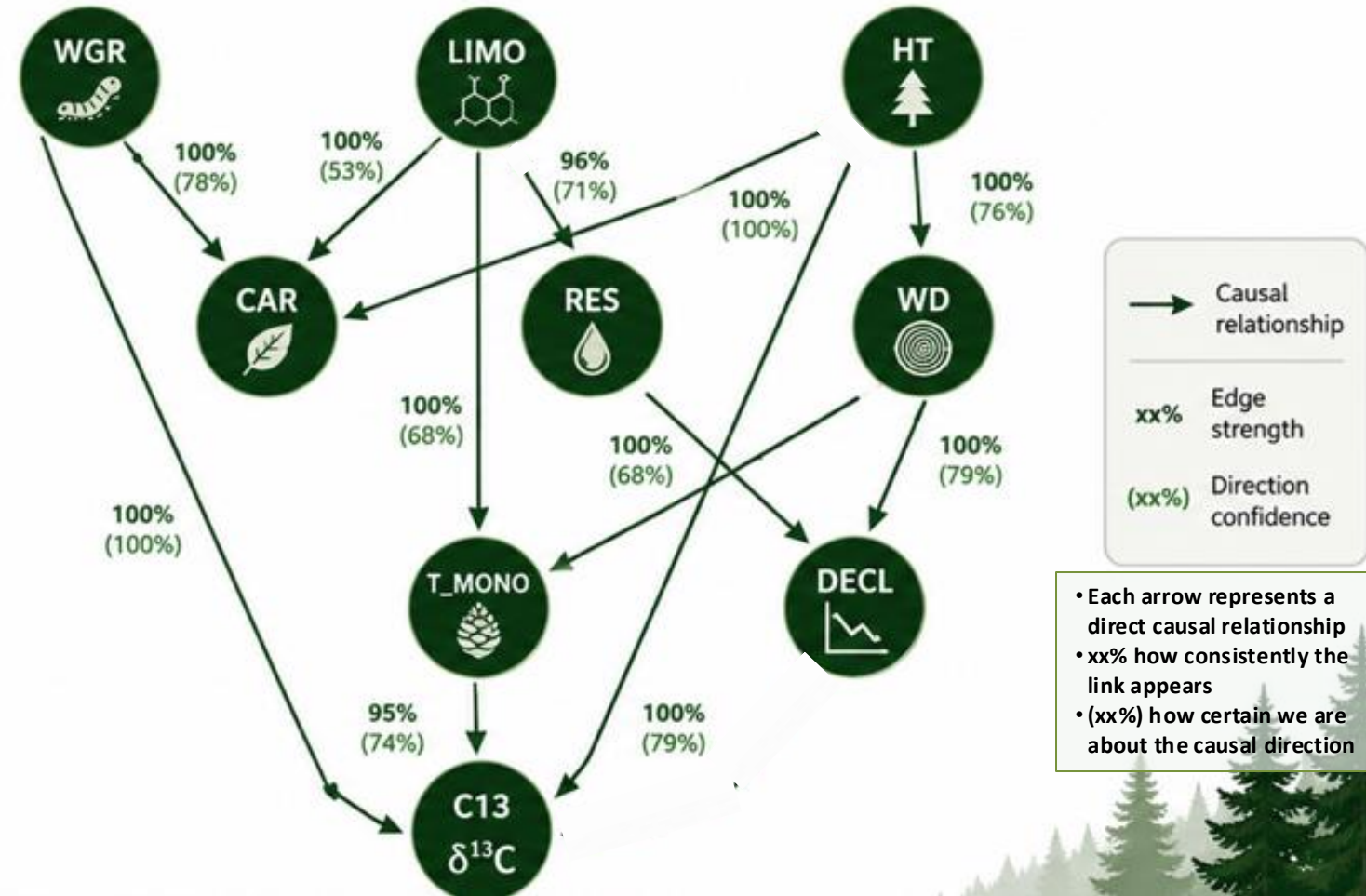
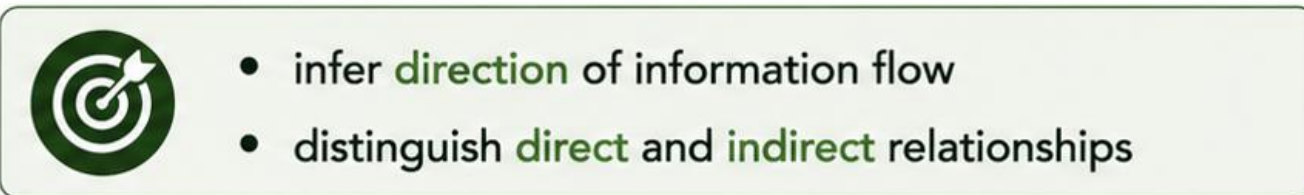
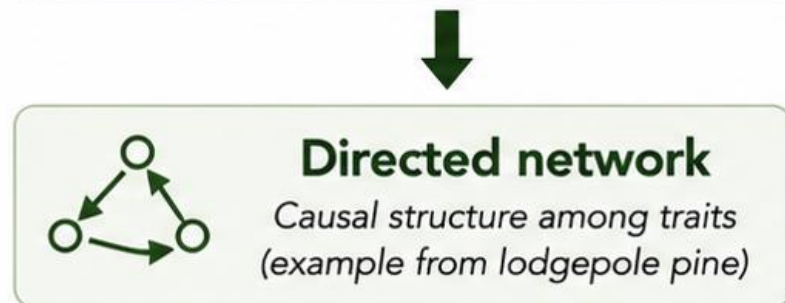
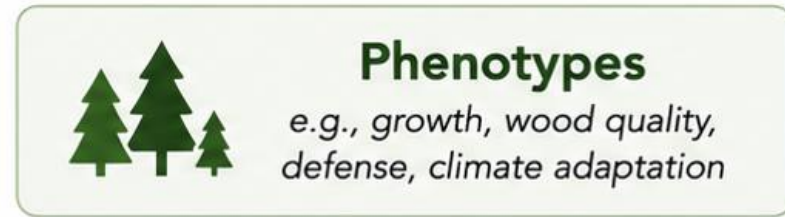
We need models that can **discover and quantify causal relationships among traits**.

Bayesian Networks & Structural Equation Models

Multi-trait models tell us that traits are related. $\longrightarrow$ Causal models try to explain **why** they are related.

Bayesian network

Cappa et al. (G3, 2025) – Lodgepole pine



Structural Equation Models improve prediction

Cappa et al. (G3, 2025) – Lodgepole pine

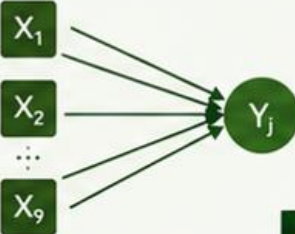
Bayesian Network



Learns the causal structure among traits

↓

Structural Equation Model (SEM)



Genetic effects reflect the direct impacts acting specifically on each trait

↓

$$y_i = \Lambda y_i + X_i' \beta^* + a_i^* + e_i^*$$

Λ is a 9×9 lower-triangular matrix of structural equation coefficients (λ).

SEM outperforms traditional multi-trait (MTM) models



↓ Residual variance
Average reduction (MTM → SEM)
8 – 21%

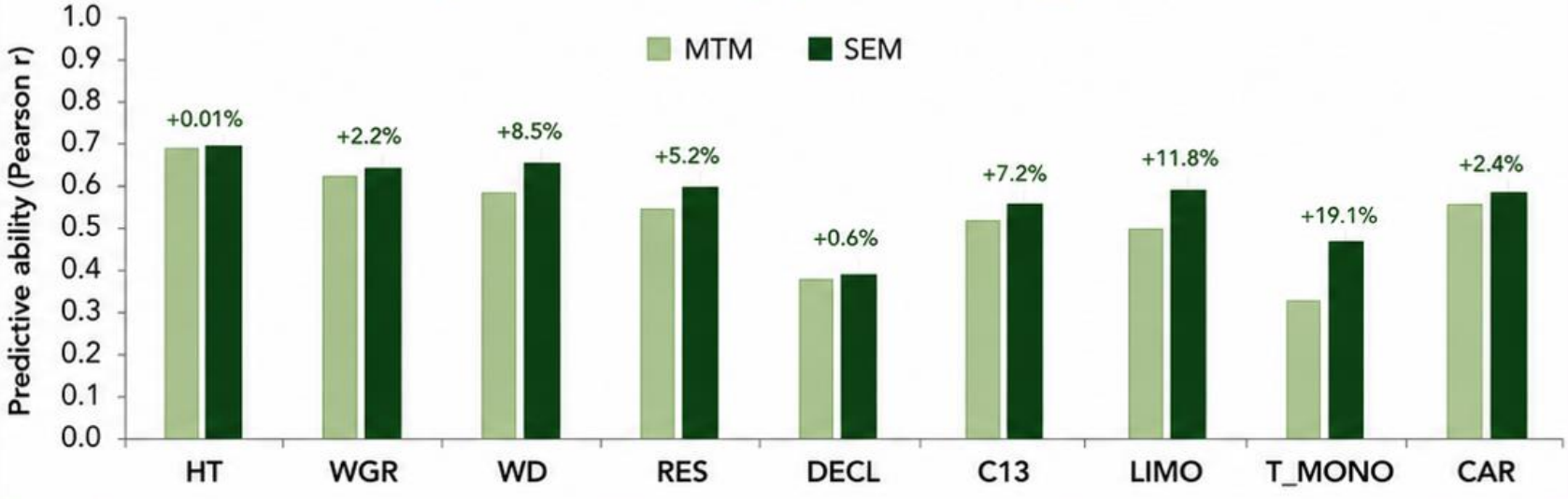


↑ Heritability
Average increase (MTM → SEM)
9 – 39%



↑ Predictive ability
Average gain in accuracy (MTM → SEM)
6%

Example: Predictive ability by trait (GBLUP approach)



Trait	MTM (Pearson r)	SEM (Pearson r)	Gain (%)
HT	0.68	0.69	+0.01%
WGR	0.62	0.64	+2.2%
WD	0.58	0.65	+8.5%
RES	0.54	0.59	+5.2%
DECL	0.38	0.39	+0.6%
C13	0.51	0.55	+7.2%
LIMO	0.49	0.59	+11.8%
T_MONO	0.32	0.48	+19.1%
CAR	0.55	0.58	+2.4%



Main message

Accounting for causal relationships produces biologically more realistic breeding value predictions.



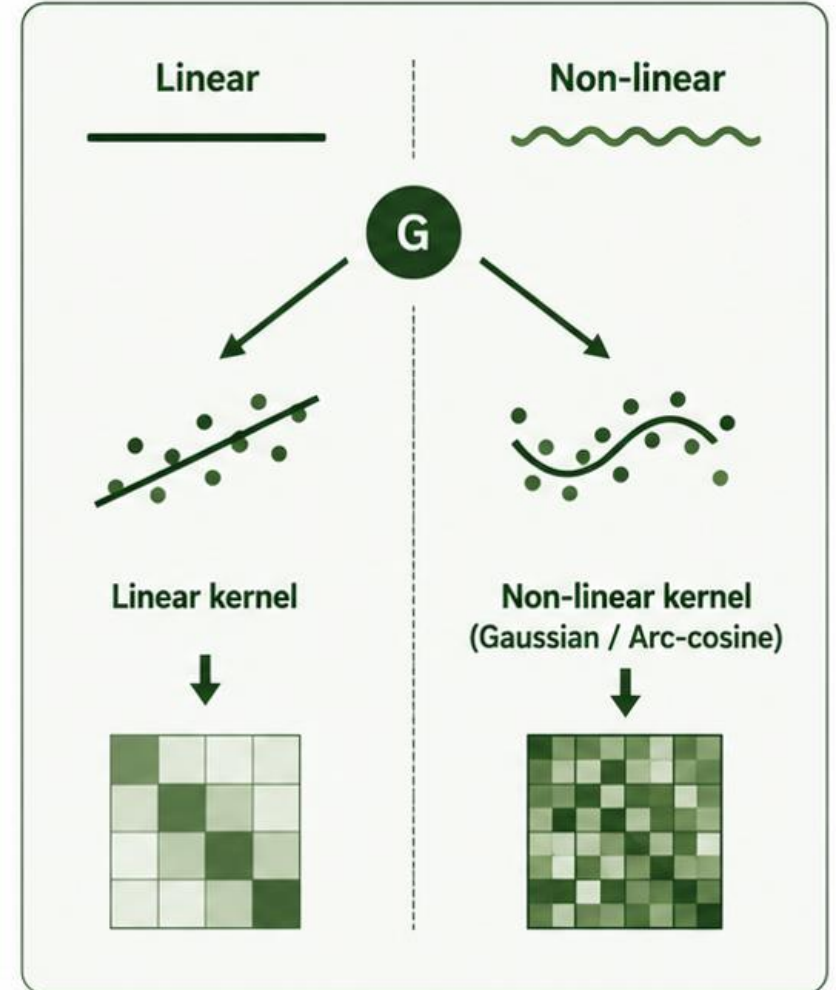
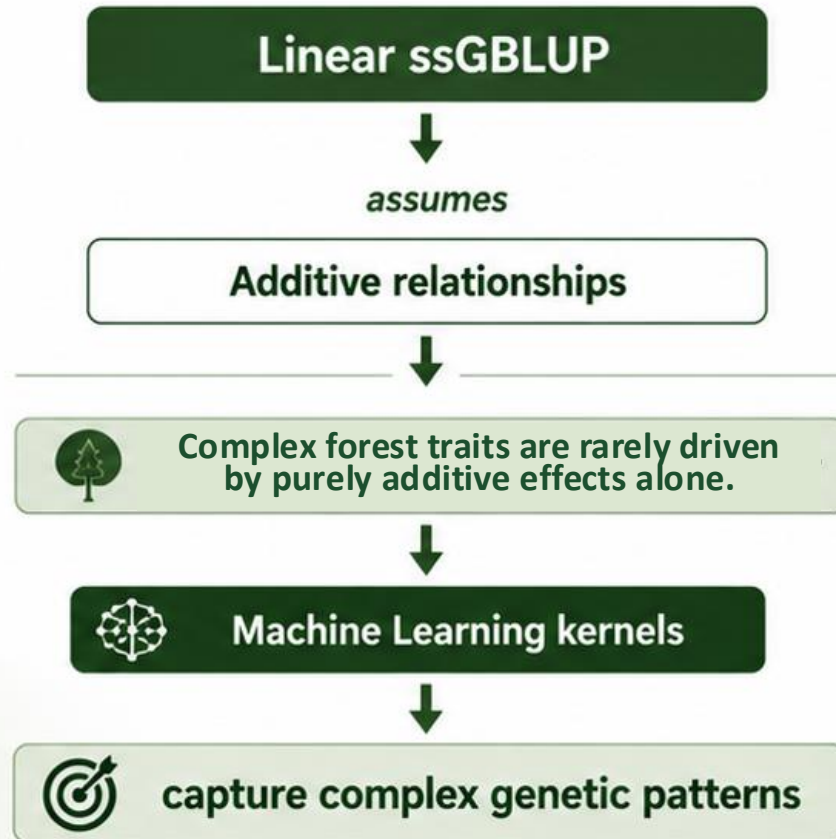
Example: Lodgepole pine

392 trees, 25,099 SNPs
9 traits (growth, defense and climate adaption)

PART III: Machine Learning for Genomic Prediction

The seventh limitation: beyond linear models

Cappa et al. (TGG, 2026) – White spruce



Can Machine Learning kernels improve genomic prediction by capturing non-linear genetic architectures?

Machine learning kernels improve genomic prediction

Cappa et al. (TGG, 2026) – White spruce

White spruce (30 traits) – Single-step GBLUP models

KERNELS EVALUATED



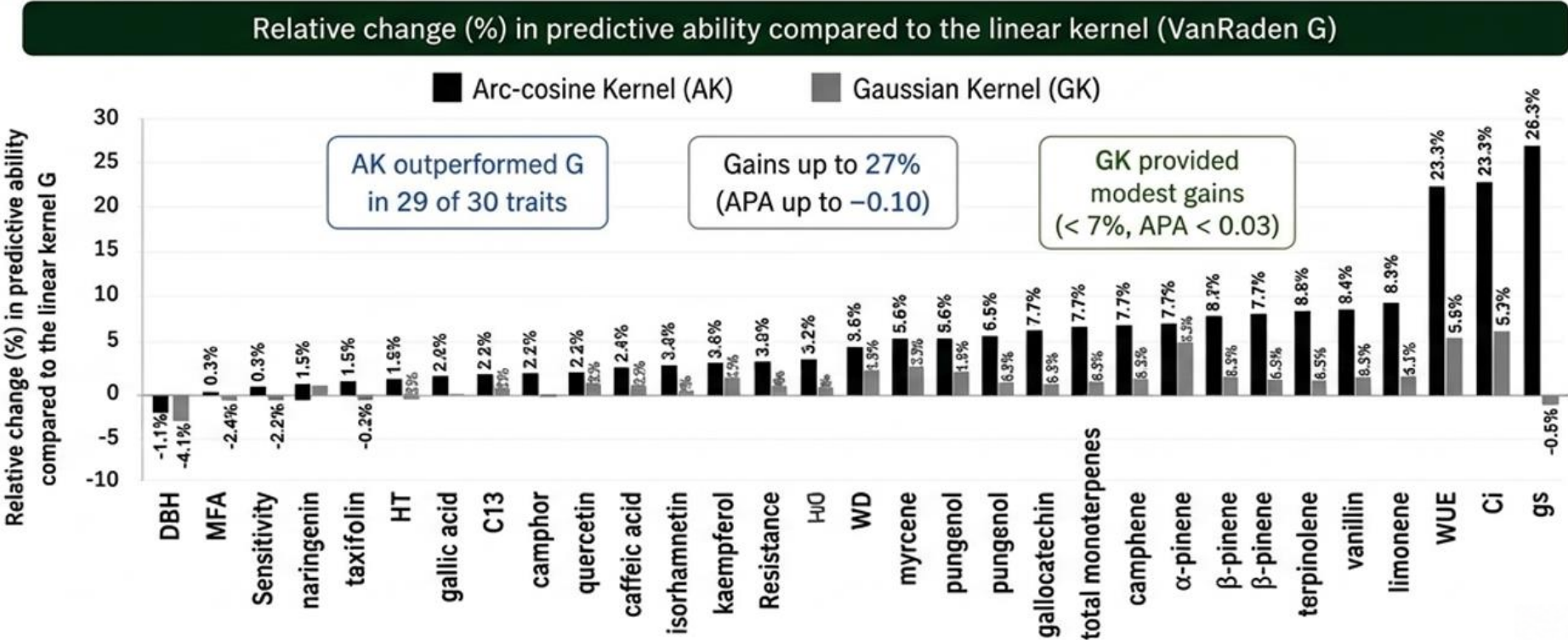
VanRaden (G)
Linear kernel
Captures additive genetic relationships.



Gaussian Kernel (GK)
Non-linear kernel
Models similarity that decays with genetic distance.



Arc-cosine Kernel (AK)
Deep non-linear kernel
Captures complex, higher-order non-linear interactions.



When connecting these prediction gains to underlying genetic architectures, distinct trends emerge across trait groups:



Physiological
(Climate adaptability)



Monoterpenes
(Chemical defense)



Polyphenols
(Chemical defense)



Productivity & Wood
(Growth & quality)



Key message

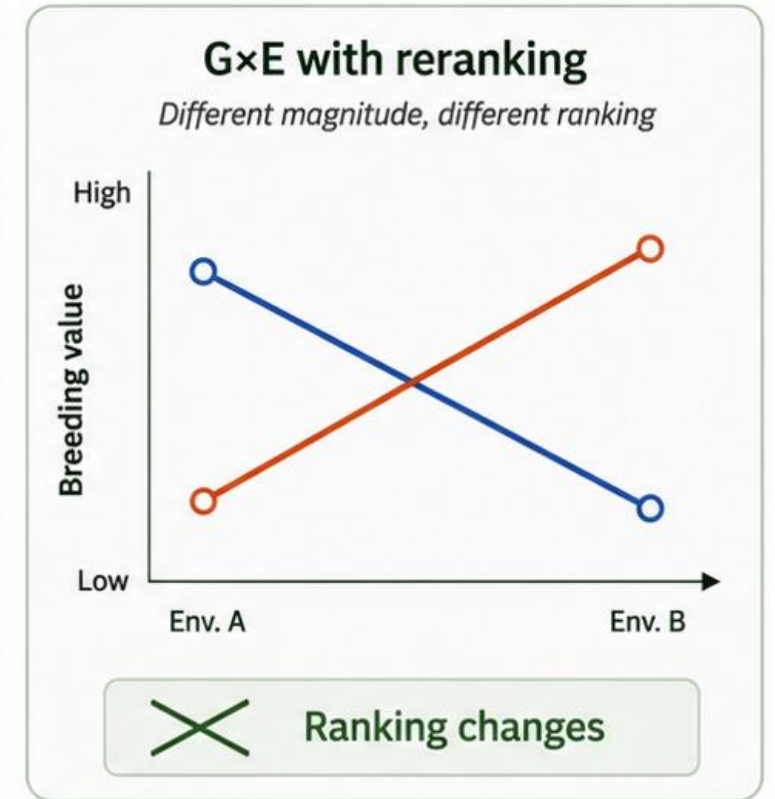
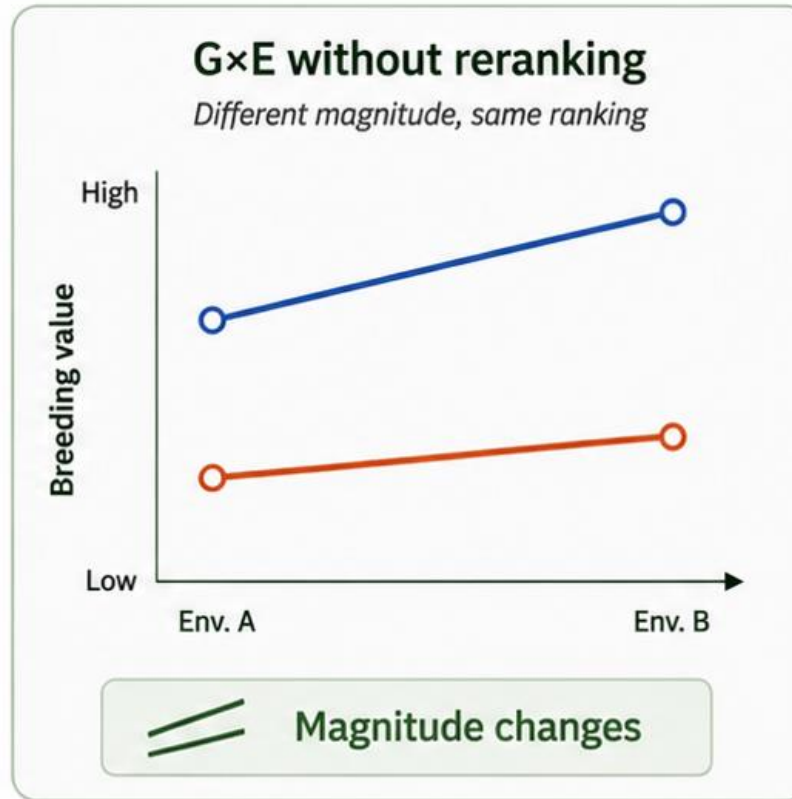
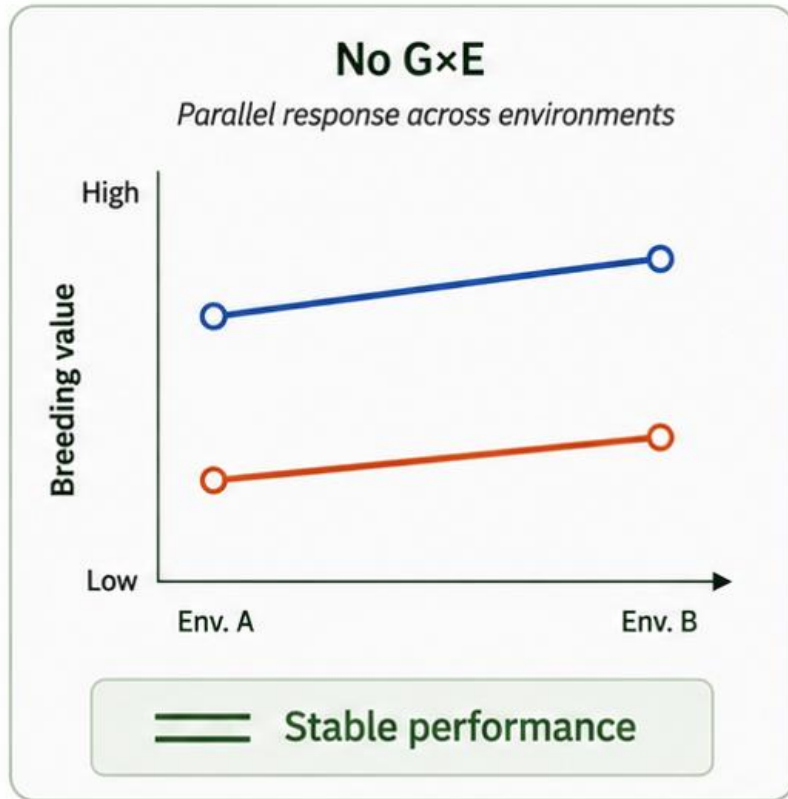
Prediction gains are **context-dependent** and depend on both the **genetic architecture** of the trait and the **choice of kernel**.



PART IV: Prediction Across Environments

Prediction across environments

Genetic merit is not always the same across environments



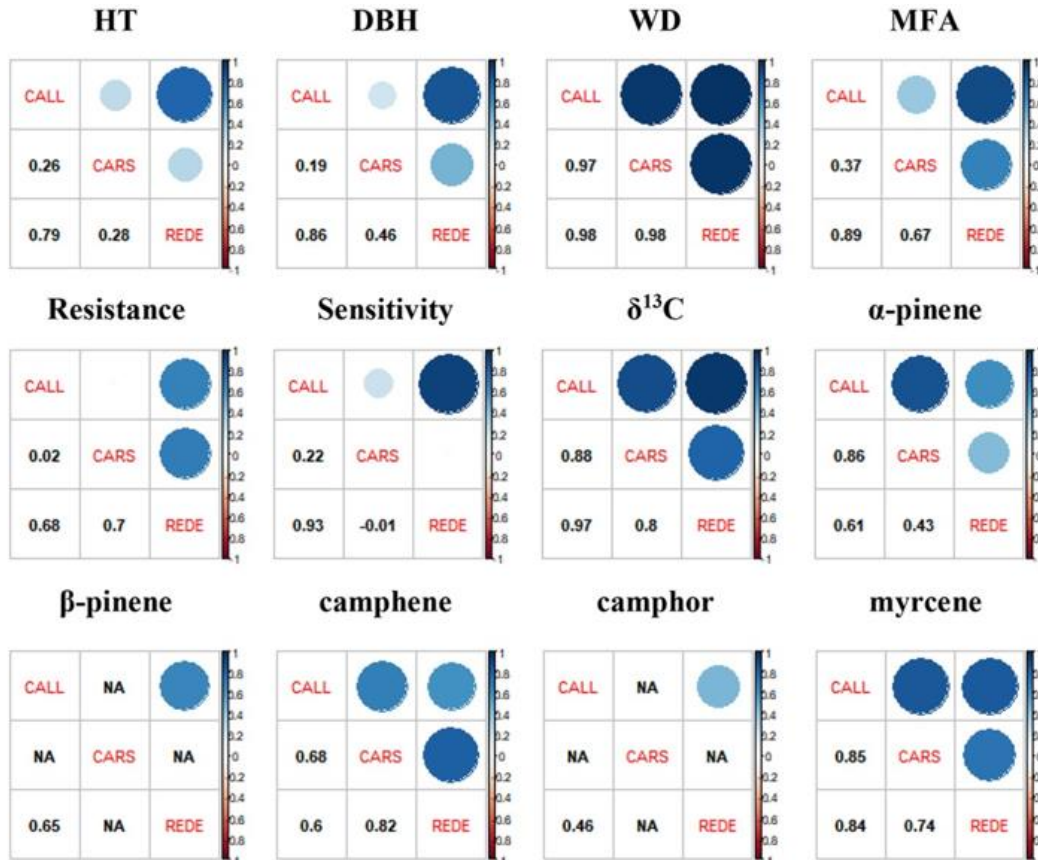
—○— Genotype 1 —○— Genotype 2

Measuring G×E using genetic correlations

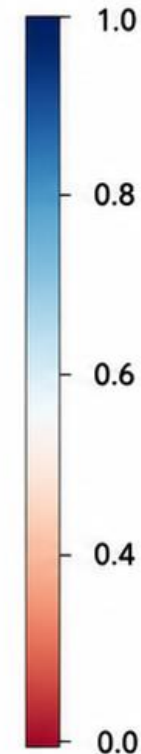
Cappa et al. (PLOS One, 2022) – White spruce



Estimated genetic correlations between sites for 12 (15) traits (White spruce population).



Genetic correlation (r_g)



Stable ranking

Mild G×E

Moderate G×E

Strong G×E

High r_g

Environment A

Environment B



Same ranking

Low r_g

Environment A

Environment B



Ranking changes

Upper diagonal shown. Circle color and size reflect the strength of genetic correlation (r_g).

NA = not estimated at the CARS site due to insufficient phenotypic data.

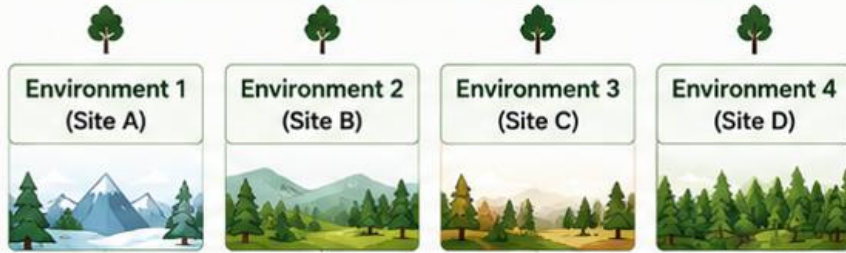


Key message

Genetic correlations provide a direct measure of G×E.

Beyond discrete environments

Traditional multi-environment models



Discrete covariance matrix (example)

	A	B	C	D
A	1.00	–	–	–
B	0.82	1.00	–	–
C	0.55	0.63	1.00	–
D	0.31	0.44	0.68	1.00

Finite number of environments

We lose valuable information, and we can only estimate genetic relationships between tested locations.

From
discrete
environments



to
continuous
environmental
variation

Continuous environmental gradients

- Temperature
- Precipitation
- Water deficit
- Altitude
- Soil properties (e.g., texture, fertility, pH)



Random regression & covariance functions allow breeding values and covariances to change continuously across environmental gradients.



Key message

Genetic prediction should follow environmental gradients,
not site categories.

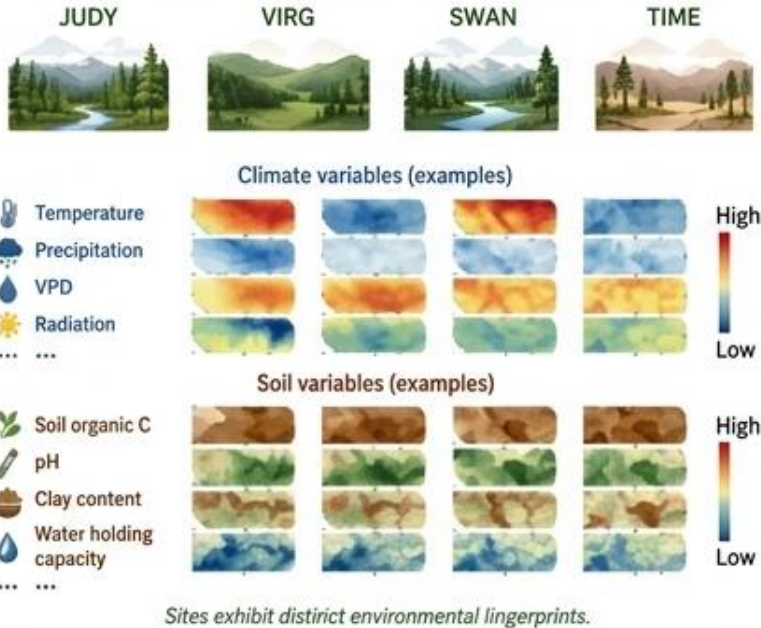


Explaining G×E using environmental covariates

Cappa et al. (In preparation, 2026) – Lodgepole pine

Using climate and soil covariates to model environmental similarity and explain site variance.

1. Environments, different locations, different conditions



2. Environmental covariates

Quantify environments using measurable descriptors

Climate covariates

- Temperature (mean, min, max)
- Precipitation
- Vapor pressure esect (VPD)
- Radiation
- Growing degree days
- ...

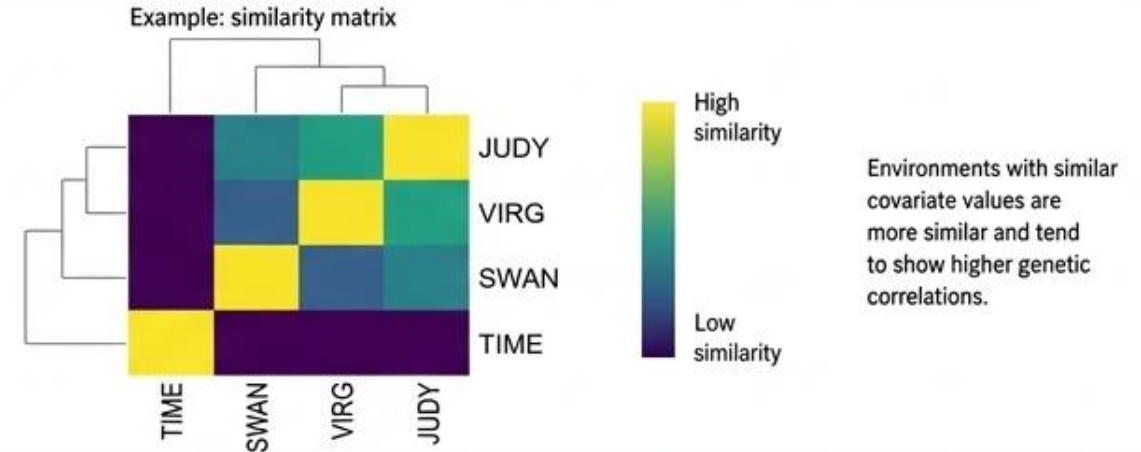
Soil covariates

- Soil organic carbon
- pH
- Soil texture (tand, clay)
- Water holding capacity
- Soil fertility
- ...

From
descriptors
to
environmental
similarity



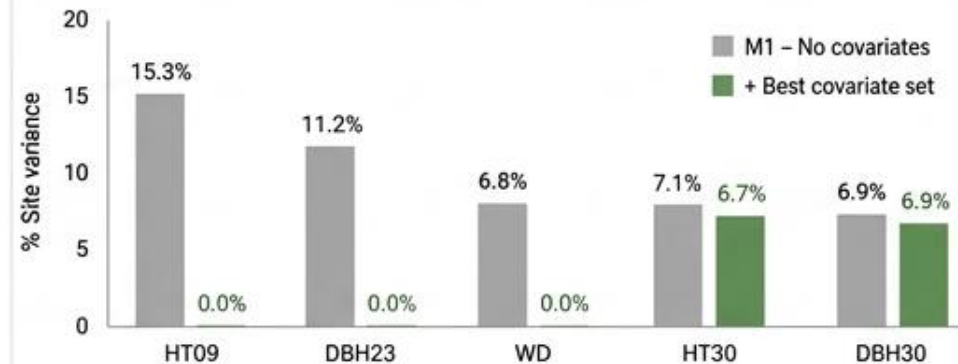
3. Environmental similarity among sites



4. Environmental covariates explain site effects

Adding covariates reduces the (co)variance attributed to sites

Examples: % of phenotypic variance due to site



Examples



HT09
Site effect reduced from 15.32% to 0.00% with soil covariates.



WD
Climate covariates reduced site effect to 0.00%.



HT30 / DBH30
Site variance remained virtually unchanged.



Key message

Environmental covariates capture environmental differences among sites and reducing the site effect.



Different traits are influenced by different environmental drivers.

Predicting breeding values across environmental gradients

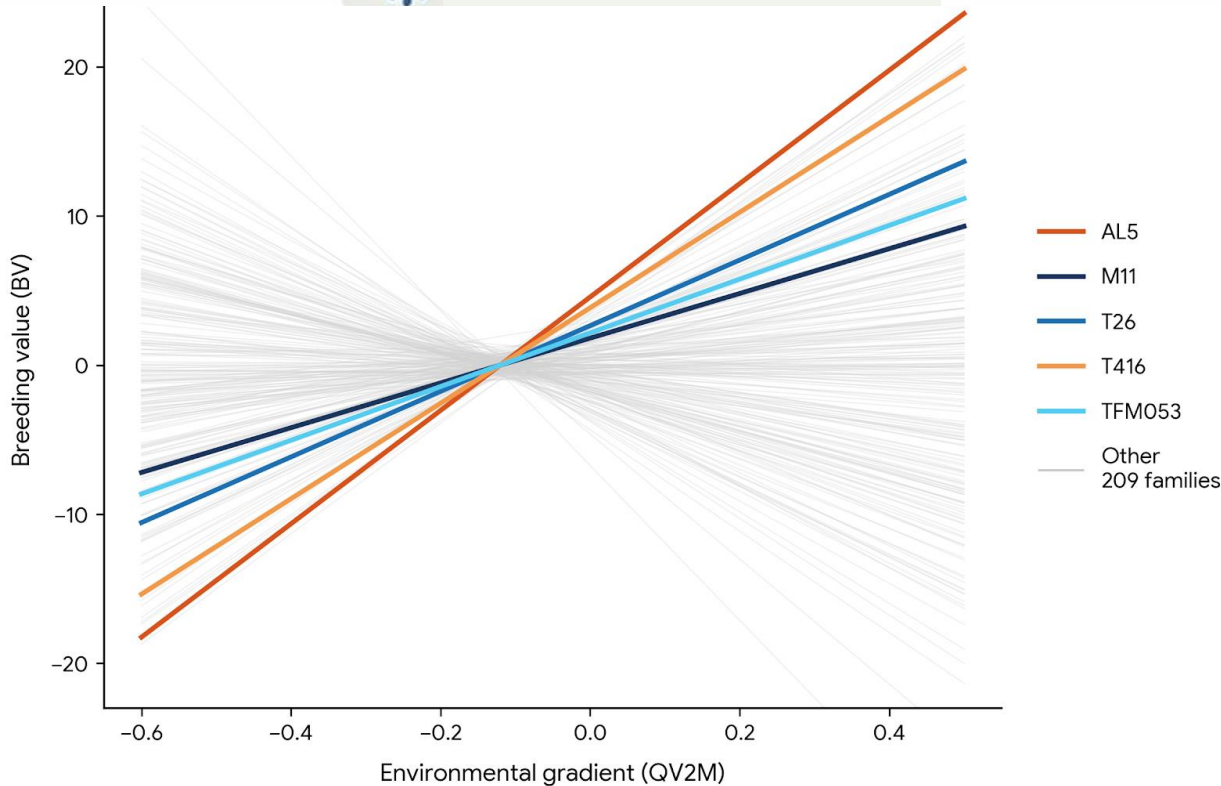
Belaber et al. (In preparation, 2026) – *Pinus taeda*

Reaction norm models describe how breeding values change along continuous environment gradients

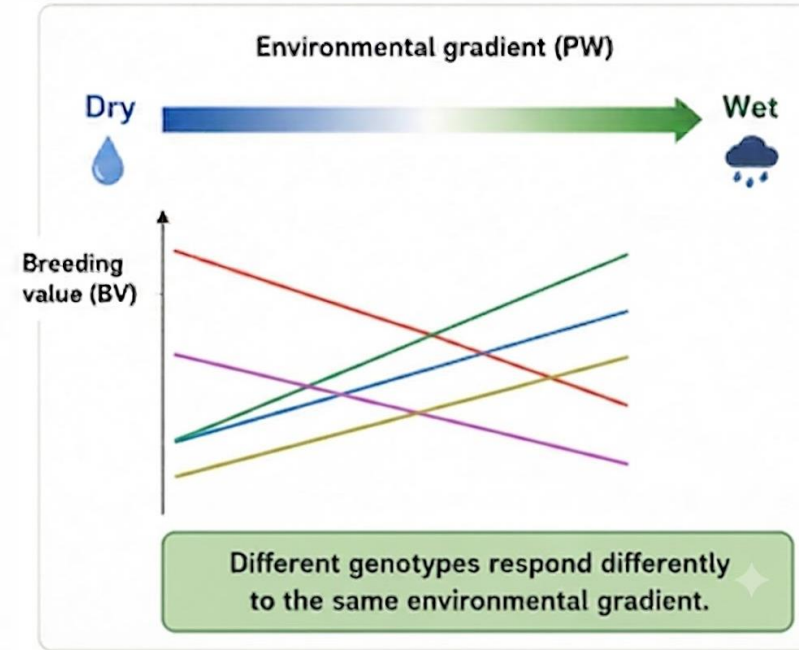
Reaction norm for 214 families along the specific humidity at 2 m. (QV2M) gradient



QV2M - Specific humidity



What do reaction norms represent?



From discrete sites to continuous prediction

Discrete sites



Continuous prediction



Conceptually:

$$BV_i(E) = a_i + b_i E$$

Breeding values are modeled as functions of continuous environmental variation.



Key message

Reaction norms enable continuous prediction of breeding values across environmental gradients, including both observed and unobserved environments.



Beyond climate and soil: adding targeted monoterpenes

Cappa et al. (In preparation, 2026) – Lodgepole pine



Targeted metabolomics

Biomarkers of adaptation and G×E

- ✓ Captures genotype × environment responses
- ✓ Adds biological information beyond climate and soil
- ✓ Targets specific metabolites linked to plant response



Climate + Soil



Targeted metabolomics



Better prediction
in unobserved
environments

Impact on predictive ability in unobserved environments

Relative change (%) in predictive ability vs. baseline model (M1)

Covariate set	Linear kernel				Gaussian kernel (AK, $h = 1$)				Arc-cosine kernel (AK, 1 layer)			
	JUDY	VIRG	SWAN	TIME	JUDY	VIRG	SWAN	TIME	JUDY	VIRG	SWAN	TIME
Model name												
Climate	0.14	0.17	5.81	2.32	0.18	0.10	5.94	2.47	3.38	0.21	5.77	2.32
Soil	0.07	0.07	5.60	2.44	0.14	0.00	2.74	2.44	0.07	0.17	5.56	2.44
Targeted monoterpenes	0.14	0.17	5.56	2.32	0.04	0.10	5.52	2.47	0.21	0.10	7.85	2.44



Largest improvement: **+7.85%**

SWAN × targeted monoterpenes × AK kernel



Targeted metabolomics adds **biological information** that climate and soil alone cannot capture.



Climate

+



Soil

+



Biology

→



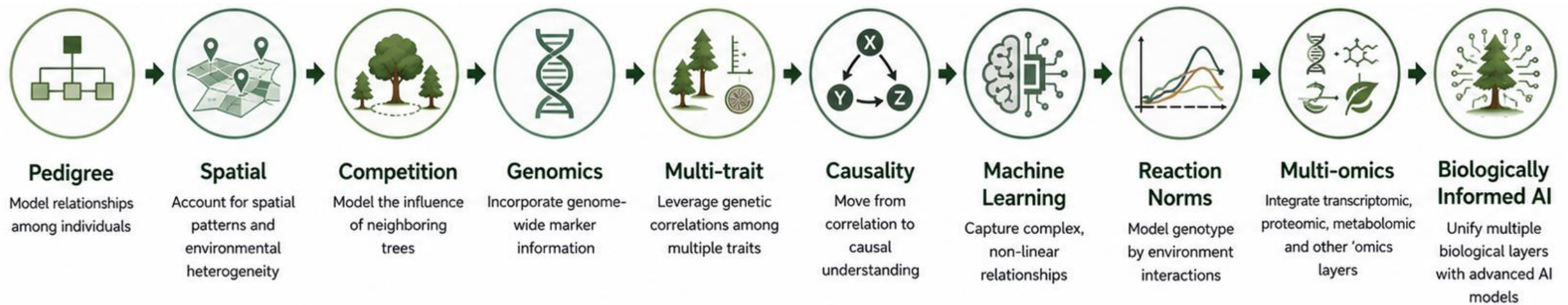
Better prediction

PART IV: Toward Biologically Informed Prediction Models

Looking ahead in forest tree breeding

The evolution: 20 years of increasing biological realism

Over the last twenty years, we have progressively incorporated new sources of biological information into prediction models.



Increasing Biological Representation



All new methods share the same goal:
building models that better capture real biology.

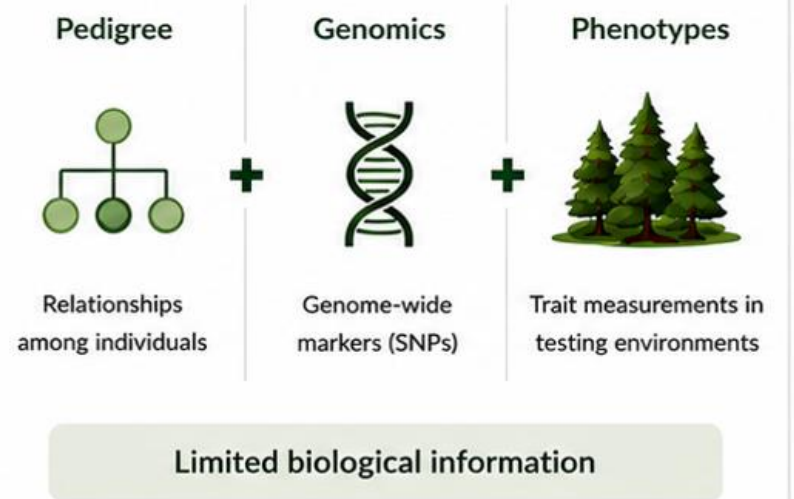
From statistical prediction to biologically informed prediction



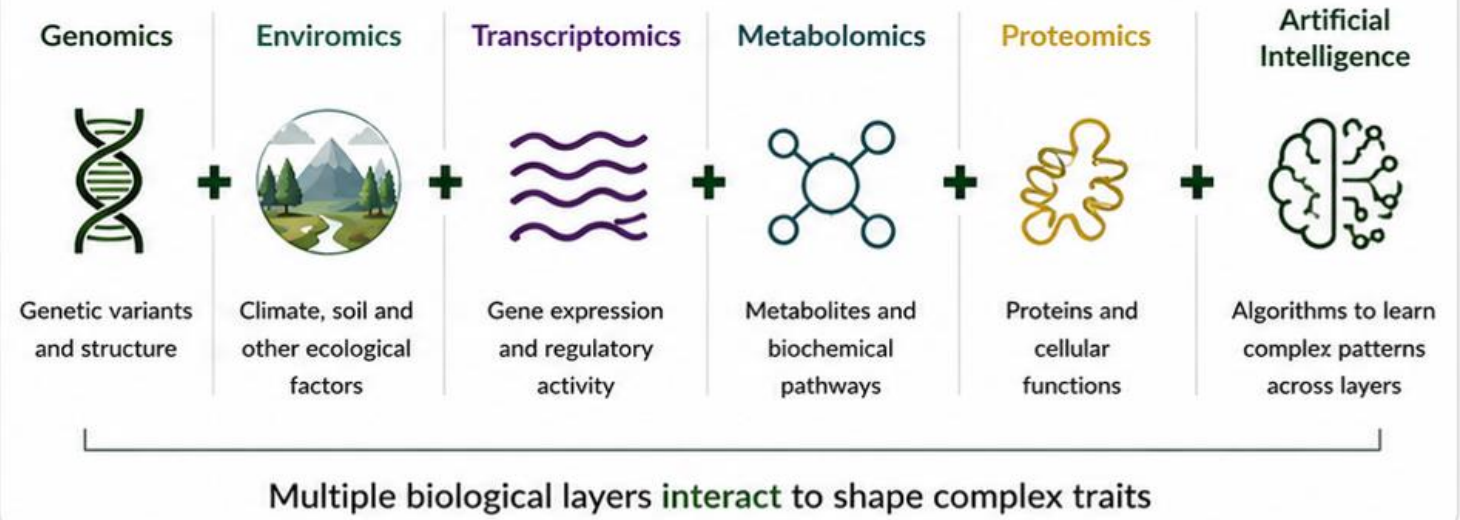
What's next?

Not just: Adding more data or complex statistical models
The real goal: Integrating biological data into a unified predictive framework

Statistical genomics-centered prediction approach



Multi-omics integrated biologically informed prediction approach



Better biological representation



Better prediction



Better breeding decisions



The future of forest tree breeding is not about building more complex statistical models.

It is about building biologically informed prediction models.

International Research Collaborations



Always open to new collaborations.

Thank you for your time!



Check out all full papers and the presentation here!



My contact

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